A_GMX: adaptive Gaussian-mixture clustering for skipper-CCD images

what every step does, the mathematics behind it, and how it performs against WADERs

comm_scripts clustering study

code of 27 October 2026

Code: `clustering_tests/adaptive.py` (`adaptive_cluster(..., engine="gmx")`), `clustering_tests/gmx.py` (`general_cluster`), `clustering_tests/agent_methods.py` (background), `clustering_tests/fe_candidates.py` (few-electron candidate catalogue); in the pipeline `run_AGMX` or `run_WADERs --clusterer agmx`. The step figures are produced on a real image by `make_agmx_figures.py` (`AGMX_FIG_OUT=figures_current MUON_PHYS=1`); the performance figures by `make_agmx_perf.py`. A short version of this document is `aGMX_compact.pdf`.

Contents

1 Summary **3**

2 The image model and notation **5**

3 Step 0 — the background model **5**

4 Step A — the adaptive switch **7**

5 Step 1 — where are the deposits? **7**

5.1 The matched filter 7

5.2 Scale-space seeds 8

5.3 The Poisson check 9

6 Step 2 — fitting all deposits at once **10**

6.1 The model 10

6.2 What is maximised 10

6.3 E-step: who owns each pixel? 11

6.4 M-step: update each cloud from its share 11

6.5 Why this is the heart of A_GMX 12

7 Step 3 — residual pass and pruning **15**

8 Step 4 — which clouds are one deposit? **15**

9 Step 5 — tracks versus several X-rays **16**

10 Step 6 — energy, shape, purity **17**

11 Step 7 — few-electron events **19**

12 Step 8 — classes **20**

13 Step 9 — crowded CCDs (crowded mode) **20**

14 After A_GMX: the few-electron candidate catalogue **21**

15 The worked example **21**

16 Performance **22**

16.1 The crowded CCD: Am-241 10–30 keV constrained fit (code of 27 October) 22

16.2 Fe-55 front and back side through the full pipeline (code of 27 October) 23

16.3 Sharp X-rays next to diffuse ones (simulation, code of 27 October) 23

16.4 Few-electron events at low dark current (code of 26–27 October) 24

16.5 Real data: Fe-55 and Am-241 injections and Kα (code of 27 October) 24

16.6 The wide test: every image on the machine (code of 25 October) 24

16.7 Fe-55: energy versus cluster width, pairs, real Kα (code of 25 October) 25

16.8 Injected tracks, alphas and artifacts in real Am-241 images (code of 25 October) . 26

16.9 Run time 27

17 Quality flags and the selection for X-ray lines **27**

18 Does A_GMX supersede the other methods? **29**

19 Limits and open items **30**

20 Parameters, code, how to run **31**

20.1 Running A_GMX on WADERs data: cluster files in the WADERs format 31

A All performance tables **32**

References **35**

1 Summary

What it is. A_GMX clusters a calibrated skipper-CCD image (pixel values in electrons) without deciding pixel by pixel "in or out". It describes the image as **empty background + a sum of Gaussian charge clouds**, fits all clouds at once with the exact pixel integral of each cloud, shares every pixel's charge among the clouds in proportion to what each predicts there, and then groups clouds into physical deposits with two tests: a geometric one (is there a dip between two clouds?) and a statistical one (is this region several round X-rays, or one irregular deposit?). A switch in front of it hands images whose background is *not* dark current + read noise to WADERs-like hysteresis clustering. On electron-counting CCDs a few-electron finder with a calibrated false-alarm rate runs as well, and a candidate catalogue for rare-event searches is written after it.

Why. WADERs' fixed thresholds (10 and 30 times the pedestal sigma) cut the tails of diffuse clusters (energies up to 8 % low on average, 26 % on the worst CCD; on the back side of noisy images 3–6 %), merge X-rays closer than ∼6 px, cannot see few-electron events, and on a crowded CCD every threshold is a compromise between merging and cutting.

How it does (section 16; the column "code" gives the code version of each number):

	WADERs	A_GMX	code
crowded Am-241 CCD (29 images): energy scale k (1 = right)	0.984	1.0006	27 Oct
crowded CCD: Np Lβ / Lγ / γ 26.3 / Ag K significance [σ]	15.5 / 16.0 / 14.8 / 12.9	22.4 / 20.6 / 21.1 / 18.3	27 Oct
crowded CCD: deviance/ndf of the constrained fit	1.74	2.59	27 Oct
Fe-55 back side: Kα back − front (DM25, with g(n))	−2.9 %	−0.06 %	27 Oct
Fe-55 back-side Kα width (DM25)	138 eV	70 eV	27 Oct
sharp X-ray 3 px from a diffuse one (simulation): found / energy within 5 %	—	0.99 / 0.97	27 Oct
Fe-55 energy of injected clusters, width 0.3 → 1.8 px (median of 8 CCDs)	0.997 → 0.921	1.000 → 1.000	25 Oct
wide test, 878 image-CCDs: Fe-55 / muon / worm / 4-px pair success	0.86 / 0.59 / 0.74 / 0.11	0.93 / 0.64 / 0.79 / 0.88	25 Oct
Fe-55 energy resolution (rms of E/E_{true}), wide test	0.52 %	0.19 %	25 Oct
few-electron events at $10^{-3}\,e^-$/pix: 3-e^- found (fakes per 1.7 Mpix)	0.00	0.85 (0.07)	27 Oct
run time per image-CCD (median)	0.01 s	0.9 s (Fe-55), 62 s (Am-241)	27 Oct

The tests marked "25 Oct" were run before the background, hot-column and pixel-model changes of 25–27 October; on an 80-image-CCD subset the current code changes the real front-side Fe-55 Kα by −0.2 eV and resolves more pairs, so these numbers are expected to hold or improve; a full rerun is pending.

Recommendation (section 18). A_GMX is the general clusterer. WADERs stays the pipeline reference (and is what A_GMX itself uses on "unmodelled" images); MFD+G stays the fastest choice when only isolated X-rays matter.

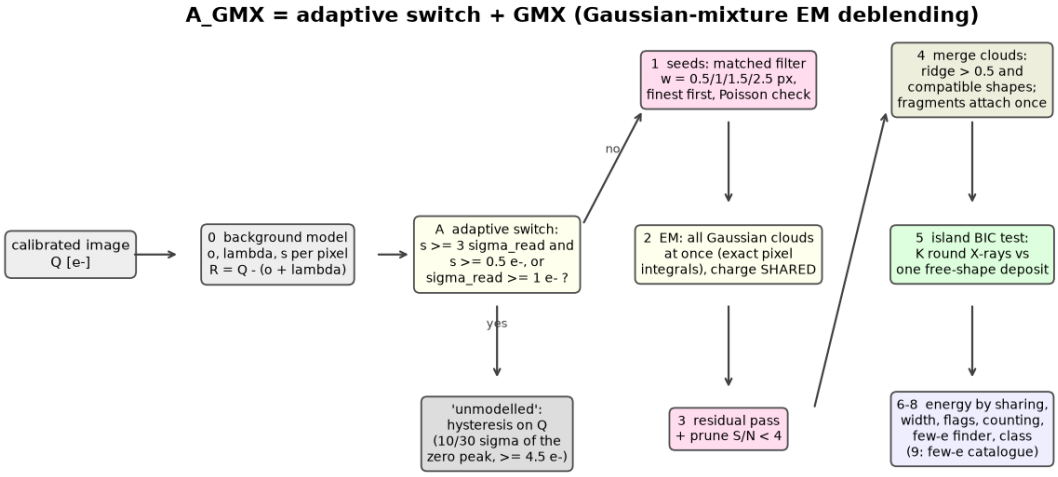


Figure 1: The steps of A_GMX (the numbers refer to the sections below).

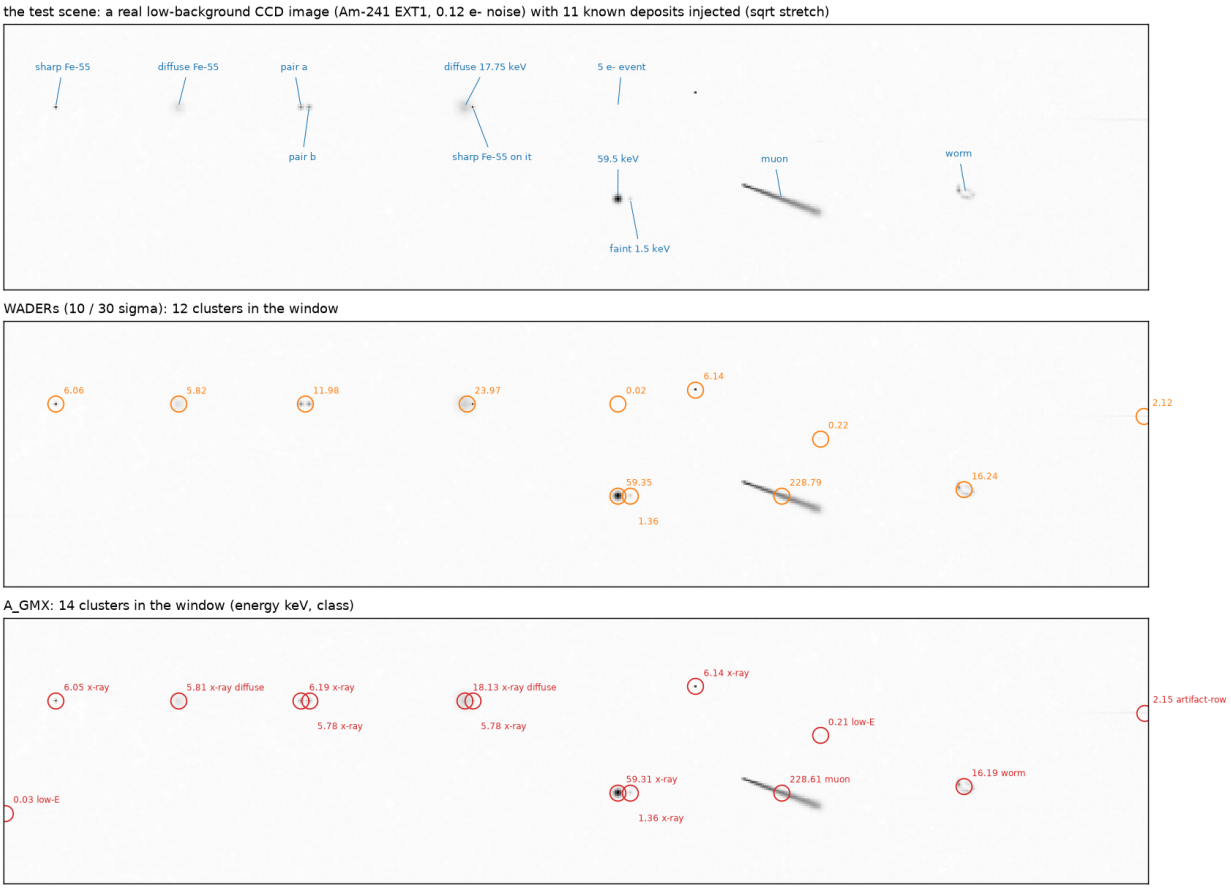


Figure 2: A real low-background CCD image (Am-241, EXT1, $0.12\,e^-$ read noise) with eleven known deposits injected (top). WADERs (middle) merges the 4-px pair into one 12-keV cluster and the sharp X-ray sitting on a diffuse one into one 24-keV cluster. A_GMX (bottom) separates both and labels the muon, the worm and the X-rays. Numbers: energies in keV.

2 The image model and notation

A calibrated image is a grid of pixel values Q_p in electrons (p = pixel). Each pixel holds

$$Q_p = o_p + n_p + \epsilon_p + D_p, \qquad n_p \sim \text{Poisson}(\lambda_p), \quad \epsilon_p \sim \mathcal{N}(0, s_p^2), \tag{1}$$

- o_p: offset (where "0 electrons" sits after calibration),
- n_p: dark charge, an integer number of electrons, Poisson with mean λ_p (e^-/pix),
- ϵ_p: read noise, Gaussian with width s_p (0.1–0.3 e^- on skipper CCDs),
- D_p: charge from particle deposits — what we want.

A deposit of charge A (electrons) whose charge cloud has covariance Σ (diffusion in the silicon: ∼0.1–0.3 px on the front side, up to ∼1.5 px on the back side) puts on average

$$D_p = A \int_{\text{pixel } p} \mathcal{N}(\mathbf{u}; \boldsymbol{\mu}, \Sigma)\, d^2u \tag{2}$$

electrons in pixel p. Energy = charge × 3.74 eV/e^-. When the module's gain-versus-charge curve $g(n)$ is available, the image is converted ADU → electrons with it *before* A_GMX runs (`src/agmx_root.py`), so shared pixels are split after the nonlinearity is removed.

symbol	meaning
$R_p = Q_p - (o_p + \lambda_p)$	residual charge: ≈ 0± noise if empty, ≈ deposit charge otherwise
$V_p = \lambda_p + s_p^2$	variance of an empty pixel
$R_p^+ = \max(R_p, 0)$	the non-negative charge used as weight in the fit
$k = 1 \dots K$	clouds (components of the mixture)
$A_k,\ \boldsymbol{\mu}_k,\ \Sigma_k$	charge, centre, covariance of cloud k
$g_k(p)$	expected charge of cloud k in pixel p
$M_p = \sum_k g_k(p)$	model of the deposit charge in pixel p

3 Step 0 — the background model

Before looking for deposits, A_GMX describes the empty image (`agent_methods.background_map`). The background is estimated on the pixels that are not part of deposits, and which pixels those are is found iteratively:

Pass 1: a crude mask. With the median and the robust spread (MAD) of all active pixels, every pixel above median $+\max(4\,e^-, 5\,\text{MAD})$ is masked and the mask is grown by 5 px. Most pixels hold 0 or 1 electron, so this needs no background model; it only has to catch the obvious deposits.

Row fits. For every row, the unmasked pixels are fitted with the **Poisson ⊗ Gaussian comb**

$$P(Q) = \sum_{n=0}^{\infty} \frac{\lambda^n e^{-\lambda}}{n!}\, \mathcal{N}\big(Q;\, o + n,\, s^2\big)\,, \tag{3}$$

integer electrons smeared by read noise (figure 3, bottom). The fast fit uses the comb phase $z = \langle e^{2\pi i Q} \rangle$: the integer Poisson part drops out, so $o = \arg z/2\pi$ and $s = \sqrt{-\ln|z|/2\pi^2}$; λ is a truncated-Poisson fit of the rounded counts $n \le K$ (K a high Poisson quantile, ≥ 3), which ignores leftover cluster pixels. A row counts as resolved only if $|z|$ exceeds $\max(e^{-2\pi^2 0.45^2},\, 4/\sqrt{N})$, i.e. above the floor $\sqrt{\pi/4N}$ that N pixels of pure Gaussian noise give. Rounding counts noise above $+0.5\,e^-$ as an electron, so on these rows $\lambda \leftarrow (\lambda - p)/(1 - 2p)$ with $p = \Phi^c(0.5/s)$.

Rows without a visible comb. Where deposits cover most of a row (a charge sheet on a crowded CCD) the comb cannot be measured; when at least 5 % of the rows (and 10 rows) are resolved, such rows take o, λ, s from the nearest resolved rows (the read noise belongs to the amplifier). Only when almost no row is resolved is a pooled comb likelihood fitted over groups of 16 rows.

Smoothing. λ and s^2 are smoothed along the rows (local-linear, ± 15 rows), o is kept per row, and a 2-D correction from the mean residual in 60×128-px cells is added to λ (floor $10^{-6}\, e^-$/pix).

Pass 2: the calibrated mask. With this background, a pixel belongs to a deposit when the 3×3 box around it or the pixel alone is unlikely from dark charge and noise (Poisson tail probabilities below 3×10^{-5} and 10^{-5}; Poisson, not Gaussian, so the mask does not explode at low dark charge), grown by 4 px; the row fits are repeated with it. This mask is used only for the background; the cluster finding below starts from scratch.

This gives per-pixel maps o_p, λ_p, s_p (figure 3, top), hence

$$\text{mean}_p = o_p + \lambda_p, \qquad V_p = \lambda_p + s_p^2, \qquad R_p = Q_p - \text{mean}_p. \tag{4}$$

Read noise. $\sigma_{\text{read}} = 1.4826\times$ the MAD of the pixels with $|Q| < 0.5\, e^-$; that estimate cannot exceed $0.37\, e^-$ (a uniform spread), so above $0.3\, e^-$ the robust spread of all active pixels is used.

Hot columns. Per column, the mean of $Q - o - \lambda$ after one 5σ clip around the column median is compared with the median of the ± 10 neighbouring columns. A column whose excess d is $> 5\times$ its error and $> 0.05\, e^-$ gets $\lambda \leftarrow \lambda + d$, so the mean, the variance and the Poisson tests all include it. If more than 10 % of the columns would be flagged the image has a charge sheet, not hot columns, and nothing is flagged.

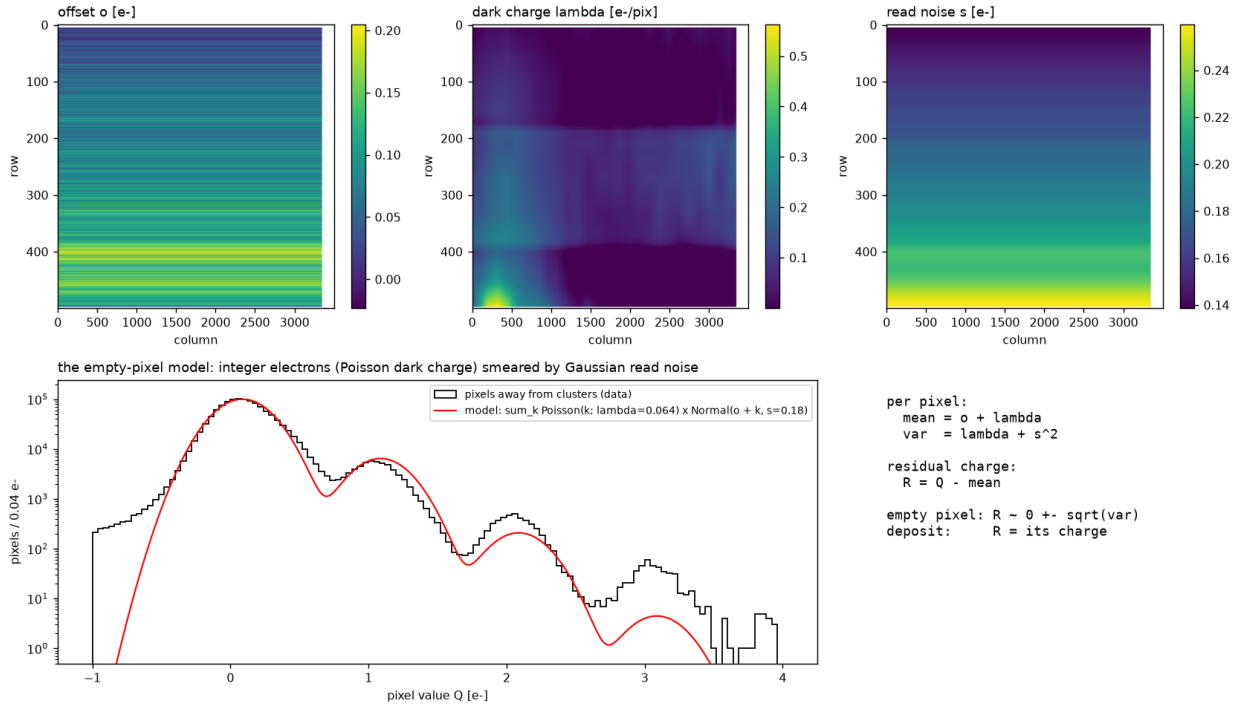


Figure 3: Top: the fitted offset, dark-charge and read-noise maps of the test image. Bottom: pixels away from clusters (black) and the comb model (red; drawn with the image's median λ and s — the fit itself is per row, which is why the data's 2- and 3-e^- peaks lie above a single-λ curve).

4 Step A — the adaptive switch

The background model can fail: on some images (the PD06/DM12 Am-241 frames) the pixels scatter far more than dark charge + read noise allow, and every model-based method then measures energies 5–10 % high. An image is **unmodelled** when

$$\boxed{\left(s \geq 0.5\,e^- \;\text{ and }\; s \geq 3\,\sigma_{\text{read}}\right) \;\text{ or }\; \sigma_{\text{read}} \geq 1\,e^-} \tag{5}$$

with s the median fitted noise. The second condition catches images that are not electron-counting at all (with a few e^- of noise per pixel the sum over a GMX footprint collects $\sqrt{N_{\text{pix}}}$ times the noise).

Unmodelled → hysteresis clustering on the image itself (the model's offset is not trusted either): with σ_z the MAD of the zero-electron peak, pixels above $\max(10\sigma_z,\, 4.5\,e^-)$ that touch (8 neighbours) form a cluster, kept if one pixel exceeds $\max(30\sigma_z,\, 9.5\,e^-)$; energy $= \sum Q_p$.

Otherwise → GMX on the whole image (steps 1–8). Every cluster also reports the **event occupancy** of its 31 × 31-px neighbourhood (fraction of pixels holding event charge; ≥ 0.3 → "crowded"); it is information only.

5 Step 1 — where are the deposits?

5.1 The matched filter

To decide whether there is a blob of width w centred at pixel c, a **matched filter** is used: a weighted sum of the residuals with the expected shape as weights. The weights are the fraction of a unit-charge Gaussian cloud of width w, centred on the centre of pixel c, that falls in each pixel $p = (i, j)$ — the exact pixel integral, separable into y and x:

$$h_w(p-c) = f(i-c_y)\, f(j-c_x), \qquad f(u) = \tfrac{1}{2}\Big[\operatorname{erf}\Big(\tfrac{u+\frac{1}{2}}{\sqrt{2}\,w}\Big) - \operatorname{erf}\Big(\tfrac{u-\frac{1}{2}}{\sqrt{2}\,w}\Big)\Big], \tag{6}$$

on a footprint of $\pm\lceil 4w+1\rceil$ px (`agent_methods._f1d`). With these weights,

$$S_w(c) = \sum_p h_w(p-c)\, R_p, \qquad \operatorname{Var}[S_w(c)] = \sum_p h_w(p-c)^2 V_p, \qquad \boxed{T_w(c) = \frac{\sum_p h_w(p-c)\, R_p}{\sqrt{\sum_p h_w(p-c)^2\, V_p}}} \tag{7}$$

T_w is the significance (in Gaussian sigmas) of a blob of width w at c. Summing many pixels with the right weights makes a deposit far more significant than any single pixel: a 5.9-keV X-ray spread over 20 pixels is invisible to a per-pixel 10σ threshold in its tails but has $T \gg 100$. Both sums are separable convolutions (`gmx._sep_conv`), computed for four widths $w \in \{0.5, 1, 1.5, 2.5\}$ px (sharp front-side X-rays to diffuse back-side ones and blobs).

Normalisation and optimality. The overall scale of h_w cancels in T_w. Because the denominator is the exact standard deviation of the numerator for an empty region, T_w is a correctly normalised significance even where the noise varies from pixel to pixel. For noise that is the same in every pixel the filter is the optimal linear detector (largest signal-to-noise for a known shape; Turin 1960). Where V_p varies across the footprint (hot columns, steep dark-charge gradients) the optimal filter would also weight each pixel by $1/V_p$, $T_w^{\text{opt}} = \sum_p h_w R_p/V_p \,/\sqrt{\sum_p h_w^2/V_p}$; the unweighted numerator used here loses a little sensitivity there but keeps the false-alarm rate correct.

At very low dark charge the noise is Poisson rather than Gaussian, and the optimal Poisson template statistic is $\sum_p n_p \ln(1 + h_w A/B_p)$ (n_p the counts, A the deposit charge, B_p the background; Ofek & Zackay 2018); this is why every seed must also pass the exact Poisson box test below. The template sits on the pixel centre only (no sub-pixel offset): the EM fit of step 2 finds the centre.

5.2 Scale-space seeds

Seeds are local maxima (5×5, over valid pixels only) with $T_w > 5$. The scales are visited **finest first**; a maximum at a coarser scale is kept only if no finer-scale seed lies within $\max(2, 1.5w)$ px; within a scale, seeds closer than that are suppressed (greedy, strongest first). Why: two X-rays 4–5 px apart give two maxima at $w = 0.5$ but one merged maximum in the middle at $w = 2.5$; finest-first keeps the two (scale-space detection, Lindeberg 1994). A lone diffuse X-ray has no fine-scale maximum (its peak is too flat), so its coarse-scale seed survives.

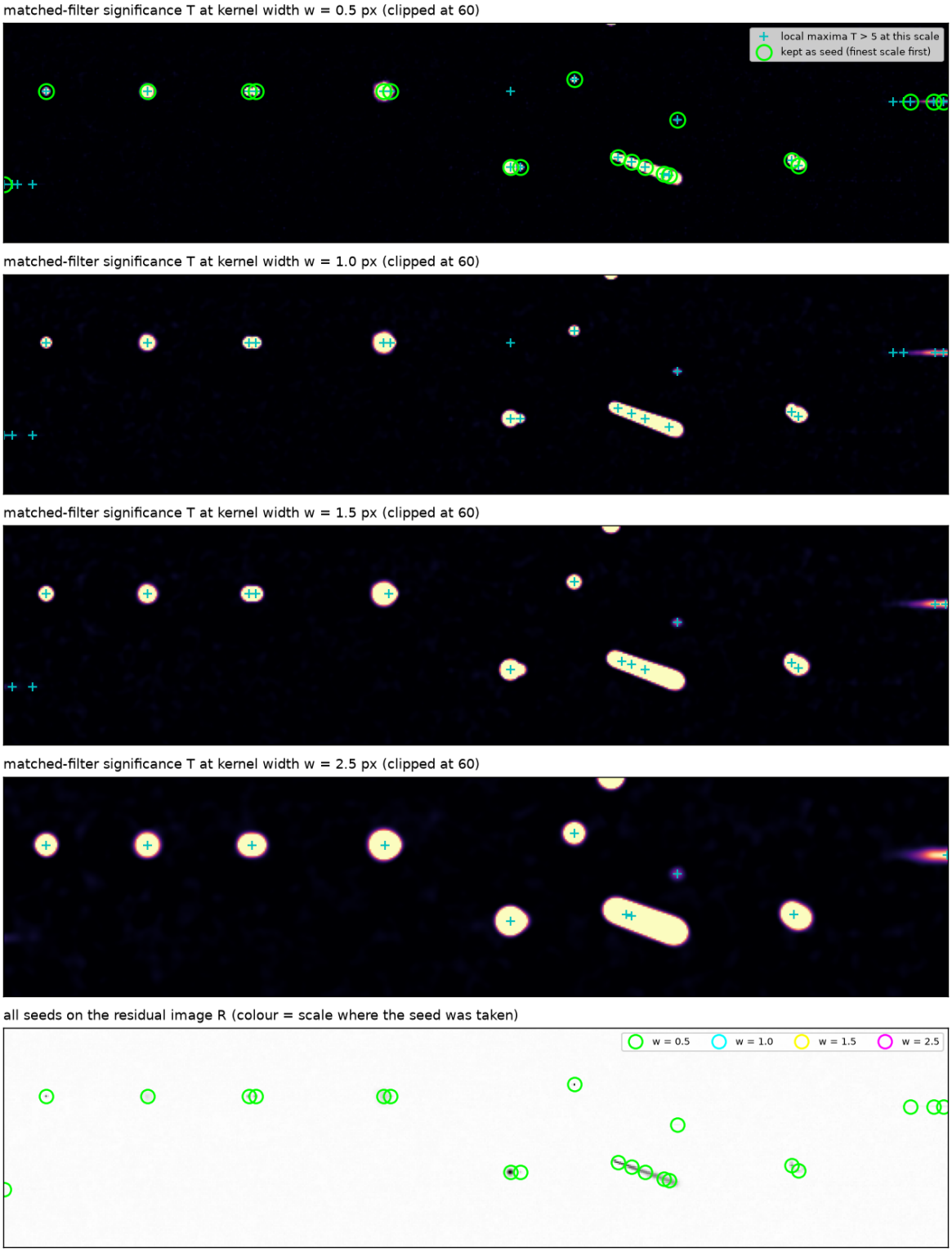


Figure 4: T_w at the four widths (clipped at 60) with the local maxima (+) and the seeds kept (circles). Bottom: all seeds on R, coloured by the scale they were taken at. The 4-px pair and the sharp X-ray on the diffuse one get two seeds each; the muon and the worm get several.

5.3 The Poisson check

T_w treats the noise as Gaussian, but dark charge is integer and Poisson: at low λ a lone box of 3 electrons is "6σ" in Gaussian terms but has probability $\sim 10^{-4}$ (figure 5). Every seed must therefore also pass an **exact counting test** on the 3×3 or the 5×5 box around it (the idea of

Li & Ma 1983 in γ-ray astronomy): with $S = \sum_{\text{box}}(Q_p - o_p)$, $\Lambda = \sum_{\text{box}} \lambda_p$, $\sigma_{\text{box}} = \sqrt{\sum_{\text{box}} s_p^2}$,

$$P(\text{Poisson}(\Lambda) \;\geq\; \lfloor S - 2\sigma_{\text{box}} + 0.5 \rfloor) < \Phi^c(5) = 2.9 \times 10^{-7}. \tag{8}$$

The $-2\sigma_{\text{box}}$ term protects against read noise pushing a box up by a fraction of an electron. On electron-counting CCDs λ here is the detection dark charge of step 7.

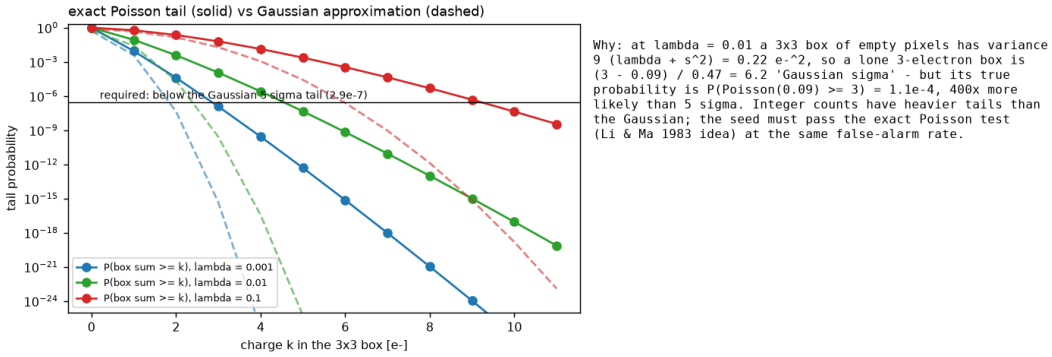


Figure 5: Tail probability of the 3×3-box charge from dark current alone: exact Poisson (solid) versus the Gaussian approximation (dashed). At low dark charge the Gaussian approximation is far too optimistic; the seed has to be below the horizontal line in the exact calculation.

Each surviving seed starts a round cloud: $\boldsymbol{\mu}$ = the seed pixel, $\Sigma = w^2\mathbf{1}$ (w = scale of the seed) and $A_0 = \max\big(R_{\text{peak}} \cdot 2\pi(w^2 + \frac{1}{12}),\, 1\big)$ (the charge of a Gaussian with that peak height).

6 Step 2 — fitting all deposits at once

6.1 The model

$$M_p = \sum_{k=1}^{K} g_k(p). \tag{9}$$

Each cloud has its own charge A_k, centre $\boldsymbol{\mu}_k$ and **full** covariance $\Sigma_k = \begin{pmatrix} \sigma_{yy} & \sigma_{xy} \\ \sigma_{xy} & \sigma_{xx} \end{pmatrix}$, so it can be round (an X-ray) or elliptical and tilted (a piece of a track). For a cloud with correlation $|\rho| = |\sigma_{xy}|/\sqrt{\sigma_{yy}\sigma_{xx}} < 0.5$ (every X-ray) the model value is the **exact integral of the Gaussian over the pixel**:

$$g_k(p) = A_k \Big[\Phi\big(\tfrac{i+\frac{1}{2}-\mu_y}{\sigma_y}\big) - \Phi\big(\tfrac{i-\frac{1}{2}-\mu_y}{\sigma_y}\big)\Big]\Big[\Phi\big(\tfrac{j+\frac{1}{2}-\mu_x}{\sigma_x}\big) - \Phi\big(\tfrac{j-\frac{1}{2}-\mu_x}{\sigma_x}\big)\Big]. \tag{10}$$

This matters for sharp clusters: a 0.2-px cloud puts almost all its charge in one or two pixels, and a Gaussian evaluated at the pixel centre would predict a peak that is too high and tails that are too low. Strongly correlated clouds (pieces of tilted tracks, broad along the track) use $A_k\,\mathcal{N}(\mathbf{x}_p; \boldsymbol{\mu}_k, \Sigma_k + \frac{1}{12}\mathbf{1})$, the Gaussian at the pixel centre with the variance of a uniform pixel added. Each cloud is evaluated only within its 4σ ellipse (at least 2 px, at most 16 px), as sparse arrays.

6.2 What is maximised

Treat the deposit charge as **counts** with mean $M_p + \varepsilon_p$, where $\varepsilon_p = \sqrt{V_p}$ is a fixed "unexplained" level per pixel (a noise component of the mixture). The Poisson log-likelihood is

$$\ln L = \sum_p \Big[R_p^+ \ln\big(M_p + \varepsilon_p\big) - \big(M_p + \varepsilon_p\big)\Big]. \tag{11}$$

Maximising it over $\{A_k, \boldsymbol{\mu}_k, \Sigma_k\}$ is a missing-data problem: if we knew which cloud each electron came from, each cloud could be fitted alone. **Expectation–maximisation** (Dempster, Laird & Rubin 1977; for Poisson images Shepp & Vardi 1982, Richardson 1972, Lucy 1974) alternates two steps. For the amplitudes with the shapes held fixed, the update $A_k \leftarrow \sum_p r_{pk} R_p^+$ is the classic ML-EM (Richardson–Lucy) step and never decreases $\ln L$; the centre and covariance updates are the weighted-moment M-step of a Gaussian mixture. With the safeguards below monotonic convergence is not guaranteed in general, but the **cluster energies**, which share the measured charge, are stable: 60 instead of 30 iterations changes X-ray cluster energies by $< 0.004\,\%$ (99th percentile). Running EM to convergence (centre steps < 0.01 px) was tested on the crowded CCD and gave no consistent gain, so a fixed number of iterations is used.

6.3 E-step: who owns each pixel?

$$\boxed{r_{pk} = \frac{g_k(p)}{M_p + \varepsilon_p}} \qquad \Big(\textstyle\sum_k r_{pk} + \frac{\varepsilon_p}{M_p+\varepsilon_p} = 1\Big) \tag{12}$$

r_{pk} is the expected fraction of pixel p's charge that came from cloud k (its **responsibility**). The noise share $\varepsilon_p/(M_p + \varepsilon_p)$ is ≈ 1 far from every cloud, so noise pixels are not handed to any cloud (figure 8, right, dotted).

6.4 M-step: update each cloud from its share

The E-step gives every cloud a share of every pixel in its footprint, $w_{pk} = r_{pk} R_p^+$ (the charge of pixel p attributed to cloud k). The M-step re-estimates each cloud from its own share alone, in four steps. Pixel p sits at integer coordinates $\mathbf{x}_p = (i_p, j_p)$ (pixel centres).

1. Charge. The cloud's charge is the total charge it was given:

$$A_k = \sum_p w_{pk}. \tag{13}$$

2. Moments of the share. The charge-weighted mean position and the charge-weighted (co)variances of the pixels of the share are

$$\bar{y}_k = \frac{1}{A_k}\sum_p w_{pk}\, i_p, \qquad \bar{x}_k = \frac{1}{A_k}\sum_p w_{pk}\, j_p, \tag{14}$$

$$v_k^{yy} = \frac{1}{A_k}\sum_p w_{pk}(i_p - \bar{y}_k)^2, \qquad v_k^{xx} = \frac{1}{A_k}\sum_p w_{pk}(j_p - \bar{x}_k)^2, \qquad v_k^{xy} = \frac{1}{A_k}\sum_p w_{pk}(i_p - \bar{y}_k)(j_p - \bar{x}_k). \tag{15}$$

These are the moments of a *pixelised* cloud (charge collected in whole pixels), not yet the moments of the physical charge cloud: pixelisation shifts the mean towards the nearest pixel centre and changes the variance, by an amount that depends on the cloud's width and on where its centre sits inside the pixel.

3. From the pixelised moments to the physical cloud. For an uncorrelated cloud ($|\rho| < 0.5$, the exact pixel model of step 2) the two axes separate. Along one axis, a Gaussian of variance σ^2 centred at offset δ from a pixel centre puts the fraction

$$\pi_i(\sigma^2, \delta) = \Phi\Big(\frac{i + \frac{1}{2} - \delta}{\sigma}\Big) - \Phi\Big(\frac{i - \frac{1}{2} - \delta}{\sigma}\Big) \tag{16}$$

of its charge in pixel i. Its pixelised mean and variance are

$$\mathcal{M}(\delta,\sigma^2)=\sum_i i\,\pi_i,\qquad \mathcal{V}(\sigma^2,\delta)=\sum_i \big(i-\mathcal{M}\big)^2\pi_i. \tag{17}$$

Two examples show why this matters: a $\sigma = 0.2\,\mathrm{px}$ cloud has $\mathcal{V} = 0.01\,\mathrm{px}^2$ when centred on a pixel ($\delta = 0$) but $0.25\,\mathrm{px}^2$ when centred on a pixel edge ($\delta = 0.5$, the charge splits between two pixels); and its pixelised mean is pulled towards the pixel centre. The M-step inverts these functions, per axis:

$$\sigma_{yy,k}=\mathcal{V}^{-1}\big(v_k^{yy}\,\big|\,\delta=\mathrm{frac}(\mu_{y,k}^{\mathrm{old}})\big),\qquad \mu_{y,k}=\lfloor\bar y_k\rceil+\mathcal{M}^{-1}\big(\bar y_k-\lfloor\bar y_k\rceil\,\big|\,\sigma_{yy,k}\big), \tag{18}$$

and the same for x ($\lfloor\cdot\rceil$: the nearest integer; frac: the distance of the previous centre from its nearest pixel centre, 0–0.5). The variance is inverted at the previous centre's sub-pixel offset, then the mean at the new variance. $\mathcal{V}$ and $\mathcal{M}$ are tabulated once (`gmx._pixvar_table`: 26 offsets $\delta \in [0, 0.5]$ × 300 values $\sigma^2 \in [10^{-3}, 30]\,\mathrm{px}^2$ on a log grid; `_pixmean_table`: 101 offsets $\in [-0.5, 0.5]$ × 120 values), summing π_i over $i = -40 \ldots 40$, made monotonic in σ^2, and inverted by linear interpolation. The covariance term $\sigma_{xy,k} = v_k^{xy}$ is kept: the exact product model of step 2 does not use it, but it decides whether the cloud is correlated ($|\rho|$) and enters the merging tests of step 4.

For a strongly correlated cloud ($|\rho| \geq 0.5$, a piece of a tilted track, modelled as a Gaussian at the pixel centre with the variance of a uniform pixel added) the inversion is simply

$$\boldsymbol{\mu}_k=(\bar y_k,\bar x_k),\qquad \Sigma_k=\begin{pmatrix} v_k^{yy}-\frac{1}{12} & v_k^{xy}\\ v_k^{xy} & v_k^{xx}-\frac{1}{12}\end{pmatrix}. \tag{19}$$

4. Safeguards. Variances are clipped to $[0.02, 25]\,\mathrm{px}^2$ and the correlation to $|\sigma_{xy}| \leq 0.95\sqrt{\sigma_{yy}\sigma_{xx}}$ (the covariance stays positive definite); a centre moves at most 1.5 px per iteration (larger steps are shortened along their direction), which stops a cloud from jumping onto a neighbour; clouds with $A_k < 10^{-3}\,e^-$ are removed. The moments are taken over the cloud's footprint (4σ ellipse, at least 2 px), so the far tails beyond it are ignored; for a Gaussian that is $< 0.1\,\%$ of the charge.

E- and M-step are repeated 30 times. The iteration is compiled (numba); the numpy version (`AGMX_NUMBA=0`) gives identical results.

6.5 Why this is the heart of A_GMX

No threshold decides which pixel belongs to which cluster. Where two deposits overlap, each pixel's charge is split in proportion to what each cloud predicts there — this is the *deblending*. Three consequences:

1. **No percolation.** In a crowded field, threshold-and-connect methods chain neighbours into one blob (or need such high thresholds that tails are cut). Sharing has no such failure mode: this is why A_GMX beats WADERs on the crowded Am-241 CCD.

2. **No charge stealing.** A sharp X-ray on the shoulder of a diffuse one keeps only its own charge (figure 8).

3. **Tails are kept.** The faint tail of a diffuse cluster is explained by its Gaussian, not cut by a threshold: energies are unbiased at every width (figure 18) and the back side of Fe-55 sits at the front-side energy.

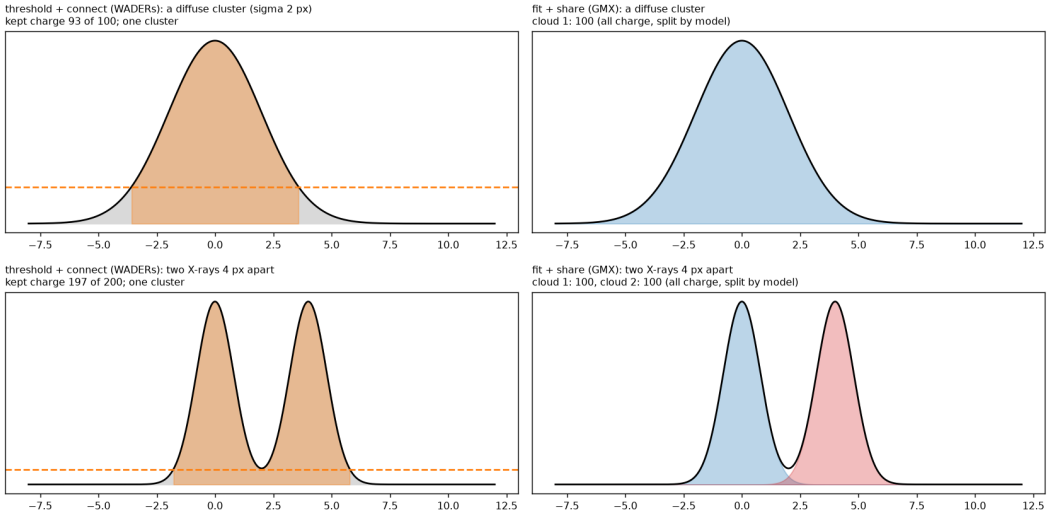


Figure 6: The difference in one dimension. Left, threshold + connect: the tails of a diffuse cluster below the threshold are lost and two X-rays 4 px apart are connected into one cluster. Right, fit + share: every electron is kept and the overlap region is split between the two clouds in proportion to their predicted charge.

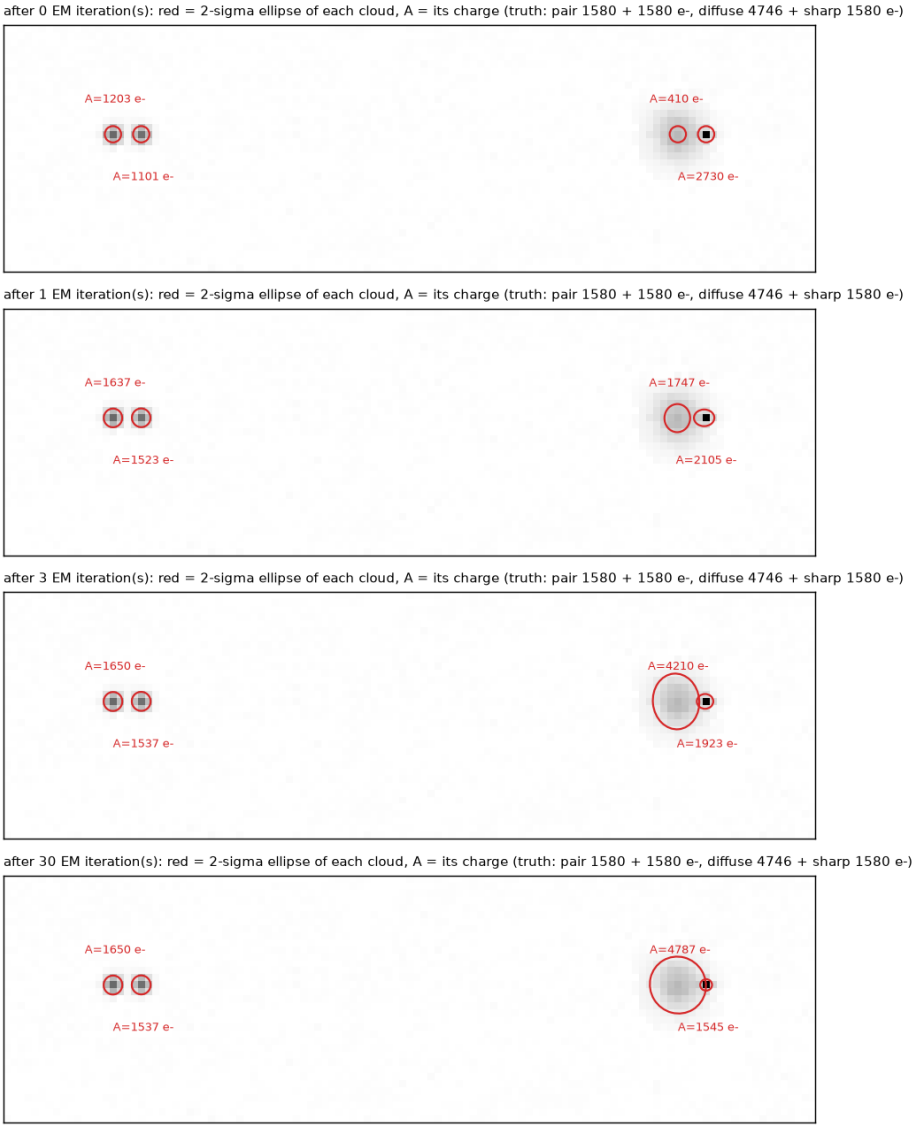


Figure 7: EM on the 4-px pair (left) and the sharp X-ray on a diffuse one (right): 2σ ellipses and charges A after 0, 1, 3 and 30 iterations. The diffuse cloud grows to its width, the sharp one shrinks onto its pixel; final charges $1650 + 1537\,e^-$ and $4787 + 1545\,e^-$ (nominal 1580 + 1580 and 4746 + 1580; the injected numbers are Poisson draws).

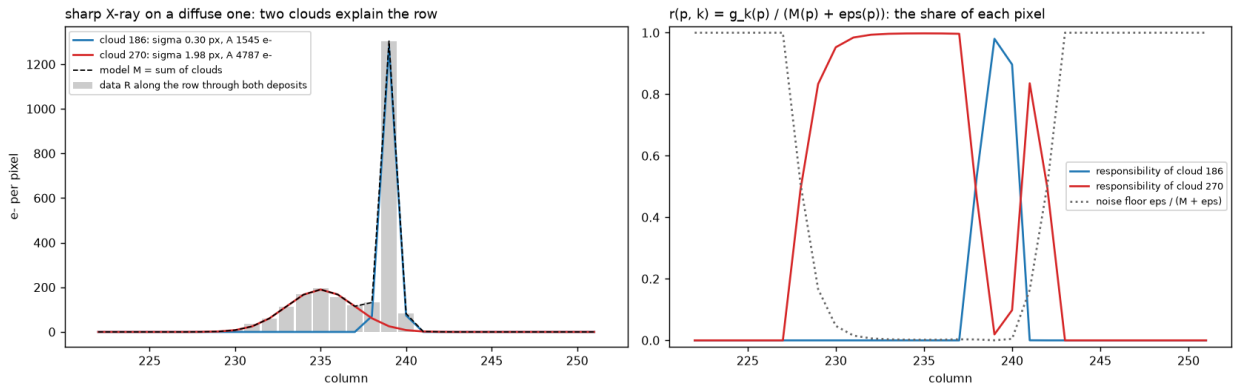


Figure 8: Left: the row through the sharp X-ray on the diffuse one — data (grey), the two clouds (σ 0.30 and 1.98 px) and their sum. Right: responsibilities; each pixel's charge is split between the two clouds, and far from both the noise share is $\sim$1.

7 Step 3 — residual pass and pruning

Residual pass. A faint deposit right next to a bright one can be hidden at the seed stage. After EM, the matched filter is run on the residual $R - M$ with the model's own Poisson variance added, $V'_p = V_p + M_p$, at the two finest scales; local maxima (3×3) with $T > 5$ that pass the Poisson check (counting CCDs) and lie more than 1 px from any cloud become new clouds, and EM runs 15 more iterations on all clouds (figure 9).

Pruning. The significance of cloud k is its charge divided by the uncertainty of that charge. For a linear fit of A_k with known profile $\pi_k(p) = g_k(p)/A_k$ and per-pixel variance v_p (read noise + dark charge + the other clouds' model, i.e. the variance if this cloud were absent), the Fisher information is $\sum_p \pi_k(p)^2/v_p$, so

$$\mathrm{S/N}_k = A_k \sqrt{\sum_p \frac{\pi_k(p)^2}{V_p + \big(M_p - g_k(p)\big)}}. \tag{20}$$

Clouds with $\mathrm{S/N} < 4$ are dropped; EM runs 5 more iterations.

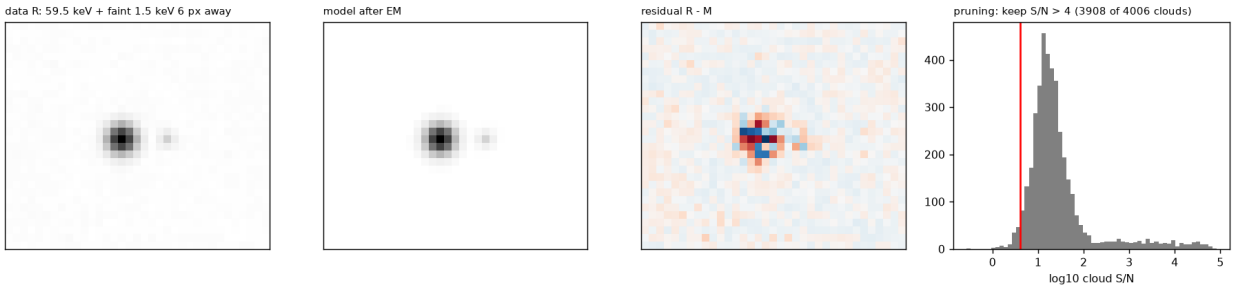


Figure 9: A 59.5-keV X-ray with a faint 1.5-keV X-ray 6 px away: data, model after the first EM, residual (the faint one stands out), and the S/N of all clouds of the image with the pruning cut (3908 of 4006 clouds kept).

8 Step 4 — which clouds are one deposit?

One deposit can need several clouds (a track needs many; a non-Gaussian X-ray may need two). Clouds are grouped by testing every pair closer than $3.5(\bar\sigma_i + \bar\sigma_j) + 1$ px ($\bar\sigma$ = geometric-mean width, at most 16 px).

Ridge (dip) test. Walk the straight line between the two centres, $\mathbf{x}(t) = \boldsymbol{\mu}_i + t(\boldsymbol{\mu}_j - \boldsymbol{\mu}_i)$, $t \in [0, 1]$, and evaluate the two-cloud density $f(t) = g_i(\mathbf{x}(t)) + g_j(\mathbf{x}(t))$ at 17 points:

$$\rho_{ij} = \frac{\min_{t\in[0,1]} f(t)}{\min\big(f(0), f(1)\big)}. \tag{21}$$

$\rho \approx 1$: no valley, one connected object. $\rho \ll 1$: two separate peaks. Required: $\rho > 0.5$ (the modality criterion for merging mixture components; Ray & Lindsay 2005, Hennig 2010).

Shape test. Even without a dip, two clouds are merged only if **one** of these holds:

- similar widths: $\max(\bar\sigma_i, \bar\sigma_j)/\min(\bar\sigma_i, \bar\sigma_j) < 1.6$;
- a track: one cloud is elongated (axis ratio > 1.4) **and** its long axis points along the line joining them ($|\cos| > 0.8$);

- a shoulder: the narrower cloud is not a local maximum of f (it only shapes the flank of the broad one).

The case this protects: a **sharp X-ray on top of a diffuse one** is a maximum of its own with a very different width — kept apart.

Fragments. A cloud below $50\,e^-$ *and* below 10 % of the charge of its brightest merge partner (a few electrons in the tail of a large deposit, often fitted as a thin "track") links only to that partner. It can be part of a cluster but cannot connect two: without this rule such a fragment linked an X-ray to the big deposit next to it. Pairs that pass are linked; the connected components of the links are the clusters.

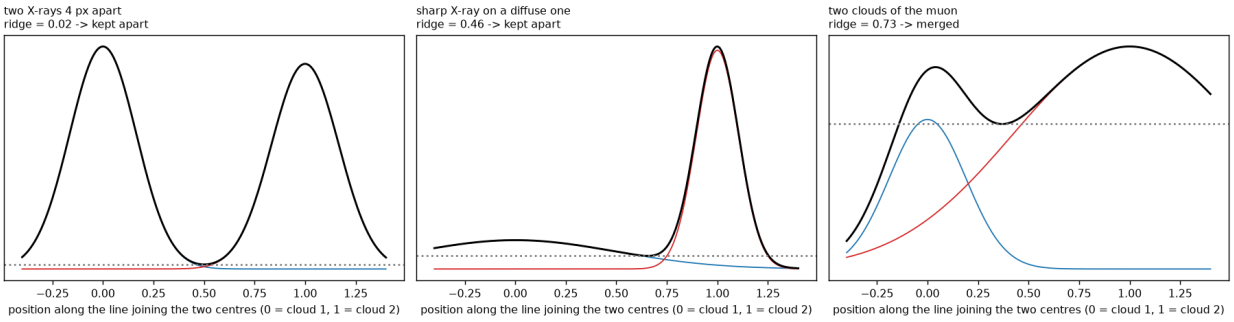


Figure 10: Two-cloud density along the line between centres. Left: two X-rays 4 px apart, deep valley ($\rho = 0.02$), kept apart. Middle: sharp X-ray on a diffuse one — $\rho = 0.46 < 0.5$, kept apart (it is also a maximum of its own with a very different width). Right: two clouds of the muon, $\rho = 0.73$, merged.

9 Step 5 — tracks versus several X-rays

Lumpy electron tracks ("worms") have real dips along them (their $\mathrm{d}E/\mathrm{d}x$ fluctuates), so the ridge test alone can split them. The **island test** asks the data directly.

Islands are connected regions (8 neighbours) of pixels above the Poisson-tail threshold of the dark charge at $\alpha = 0.01$. An island of at most 4000 px that holds $K \geq 2$ clusters (from step 4) is fitted with two hypotheses on its pixels (+2 rings):

- H_{xr}: K **separate X-rays** — one *round* Gaussian per cluster (started from the cluster's clouds, refitted by EM with $\Sigma = \sigma^2\mathbf{1}$): 4 parameters each (A, y, x, σ);
- H_{free}: **one irregular deposit** — the free-shape clouds already fitted: 6 parameters each $(A, y, x, \sigma_{yy}, \sigma_{xx}, \sigma_{xy})$, n_c clouds.

Each is scored with a weighted χ^2 and the **Bayesian information criterion** (Schwarz 1978), which charges $\ln n$ per parameter (n = pixels):

$$\chi^2_H = \sum_p \frac{\left(R_p - M_p^H\right)^2}{V_p + R_p^+ + (0.05\,R_p)^2}, \qquad \Delta\mathrm{BIC} = \left(\chi^2_{\mathrm{xr}} + 4K\ln n\right) - \left(\chi^2_{\mathrm{free}} + 6n_c\ln n\right). \tag{22}$$

The $(0.05R)^2$ term tolerates the fact that real X-rays are not perfect Gaussians (5 % model error). If $\Delta\mathrm{BIC} > 100$ — the free-shape description is so much better that it is worth its extra parameters — **the whole island becomes one deposit**. The margin of 100 is calibrated on simulated sharp-on-diffuse scenes with the exact pixel model. In crowded areas islands percolate beyond 4000 px and the test switches itself off (there, sharing does the work).

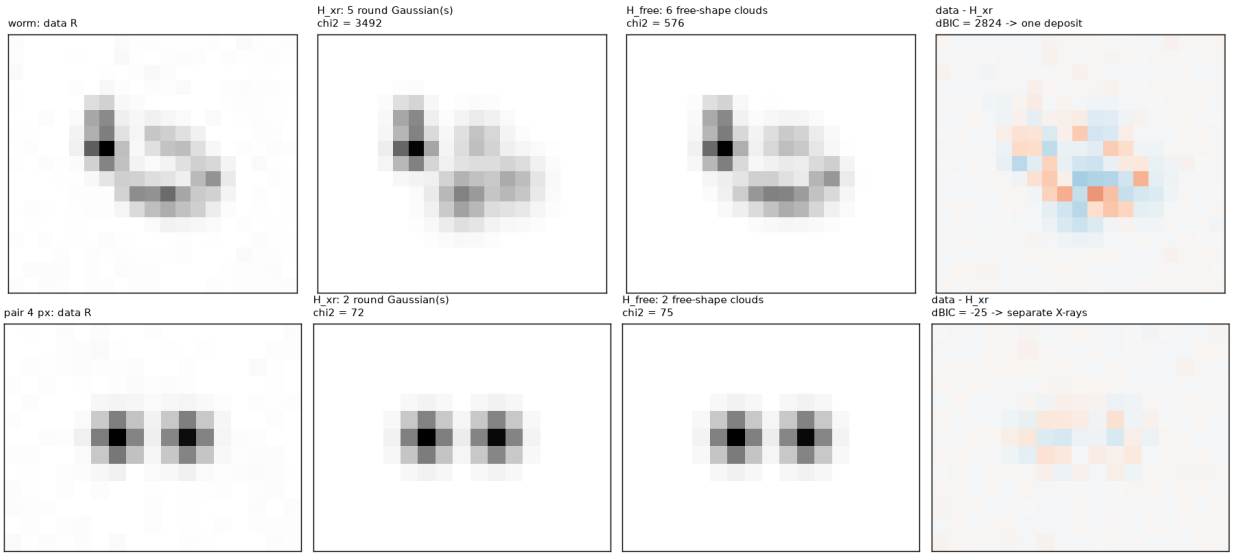


Figure 11: The island test on two islands of the test image. Top, the worm: 5 round Gaussians cannot follow the bent track (χ^2 3492, structured residual); 6 free-shape clouds can (χ^2 576); $\Delta\mathrm{BIC} = 2824 \rightarrow$ one deposit. Bottom, the 4-px pair: both descriptions fit equally well (χ^2 72 vs 75), the round one has fewer parameters, $\Delta\mathrm{BIC} = -25 \rightarrow$ two separate X-rays.

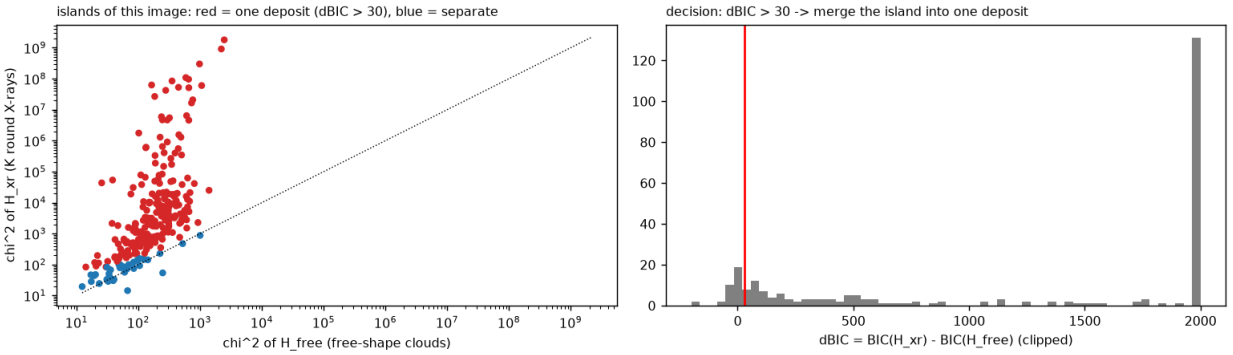


Figure 12: All islands of the test image: χ^2 of the two hypotheses (left) and ΔBIC (right). Red: one deposit (tracks, worms, irregular blobs); blue: separate X-rays.

10 Step 6 — energy, shape, purity

Energy by sharing. The cluster responsibility of pixel p is $\tilde{r}_{pc} = \sum_{k \in c} g_k(p)/M_p$ (sum over the clouds of cluster c), and

$$\boxed{E_c = \sum_p \tilde{r}_{pc}\, R_p} \qquad [e^-;\ \times 3.74\ \mathrm{eV/e^-}\ \text{for keV}] \tag{23}$$

The **measured** charge R_p is shared (not the model), so the energy does not depend on the Gaussian being exact; shared pixels are split, never counted twice.

- **Satellites.** A cluster with $E < 1\,\%$ of a neighbour's energy *and* $E < 1\,\mathrm{keV}$ whose centre lies in that neighbour's domain or within 6σ (Mahalanobis) of one of its clouds (within 20 px) is a tail fragment and is merged into it. Real X-rays ($> 1\,\mathrm{keV}$) next to a very large deposit stay their own clusters.

- **Growth.** Pixels above the Poisson-tail threshold ($\alpha = 0.01$) within 3 px of a covered pixel and connected to it, but covered by no cloud (CTI trails along the readout), are added to the nearest cluster.

Width. $\sigma_{xy}^2 = \frac{1}{2}(\mathrm{var}_y + \mathrm{var}_x) - \frac{1}{12}$, the second moments of the cluster's shared charge (unbiased to ∼1 % for Fe-55 at $\sigma \geq 0.4$ px). Below that the moment is dominated by the noise and dark charge of the footprint, so a cluster made of one cloud whose EM width is below 0.4 px reports the EM width instead:

true σ [px] (single 5.9-keV X-rays, 0.15 e^- noise)	0.2	0.3	0.4	0.6	1.0	1.5
reported σ_{xy}	0.250	0.304	0.401	0.600	0.997	1.505
$E_{\mathrm{rec}}/E_{\mathrm{true}}$	1.000	1.000	1.000	1.000	1.000	1.000

A 0.2-px cloud puts 99 % of its charge in one pixel when centred, so its width is barely measurable; either way it is far below the front/back cut at 1 px.

Significance. $E_c / \sqrt{\sum_p \tilde{r}_{pc}^2 V_p}$.

Quality flags (they never change the energy; section 17 shows what they do):

- **blend** — how much of *other* clusters' predicted charge lies under this cluster's profile, relative to its own:

$$\mathrm{blend}_c = \frac{\sum_p M_p^{(c)} M_p^{(\neg c)}}{\sum_p \left(M_p^{(c)}\right)^2}, \qquad M_p^{(c)} = \tilde{r}_{pc} M_p, \quad M_p^{(\neg c)} = M_p - M_p^{(c)}. \tag{24}$$

  0 = isolated; 0.1 = 10 % of its profile is shared with neighbours (SExtractor's deblending flag, but as a number);
- **ped** — the local pedestal: the 25th percentile of R in a ring 5–8 px around the centre (a charge sheet under the cluster);
- **peakfrac** — the share of the cluster's fitted charge in its largest cloud (merged neighbours give < 1);
- χ^2 — $\frac{1}{n_c} \sum_{p:\tilde{r}_{pc}>0.5} (R_p - M_p)^2/(V_p + M_p)$, the fit quality of its own model;
- **edge** — touches the 2-px border of the active area; **saturated** — a pixel at an ADC-limit plateau (the energy is then a lower limit).

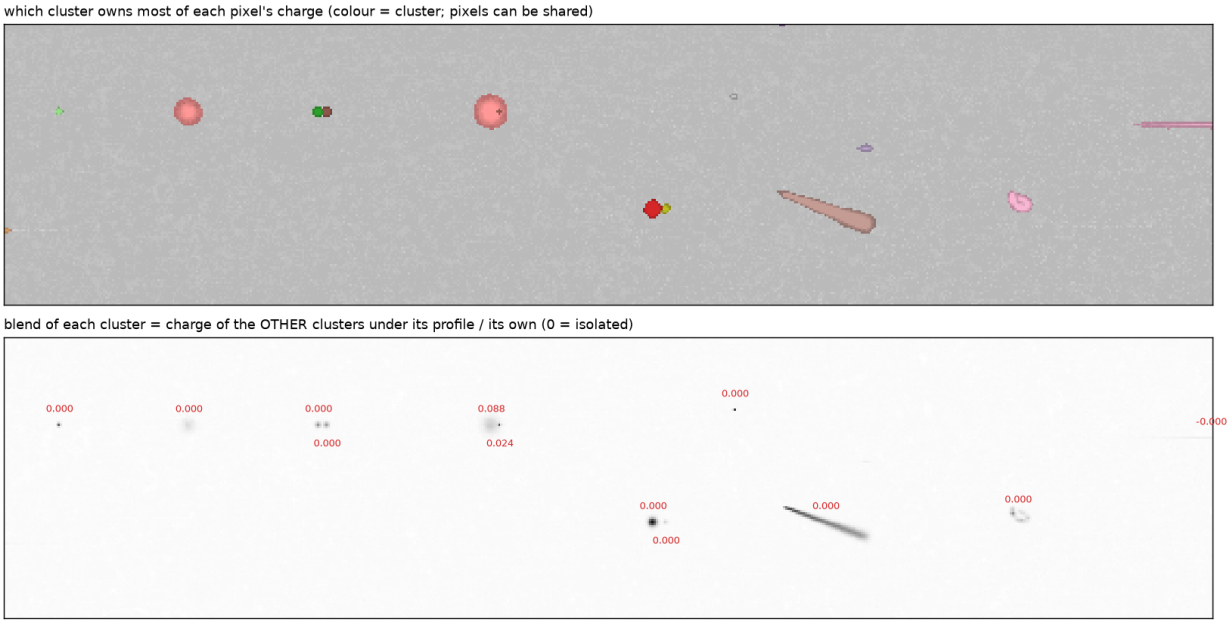


Figure 13: Top: which cluster owns most of each pixel's charge. Bottom: blend of each cluster (0 for isolated clusters; 0.088 / 0.024 for the sharp X-ray on the diffuse one, whose profiles overlap).

11 Step 7 — few-electron events

On electron-counting CCDs (median $s < 0.35\,e^-$, where rounding to whole electrons is reliable) three extra rules apply.

Detection dark charge. The fitted λ map fluctuates far below and above the truth where it is tiny (a fit cell holds only a few dark electrons). For every few-electron test the dark charge is therefore at least the one implied by the measured rate of *isolated* single-electron pixels (no other counted pixel within 2 px; real events come with neighbours), smoothed over ± 30 rows, with the rounding miscounts of the read noise included, plus the hot-column excess.

Counting energies. For clusters below $100\,e^-$, the analog sum adds the read noise of every pixel ($\sqrt{N}\,s$, $0.5\,e^-$ for 13 pixels at $0.15\,e^-$), so the charge is often off by one. Instead each pixel the cluster mostly owns ($\tilde{r} > 0.5$) is **rounded** to whole electrons and the expected dark charge is subtracted:

$$E_c^{\text{count}} = \sum_{p:\ \tilde{r}_{pc}>0.5} \Big(\text{round}(Q_p - o_p) - \lambda_p\Big). \tag{25}$$

Counting finder. An 8-connected group of at most 9 pixels with $\text{round}(Q_p - o_p) \geq 1$, not covered by any cluster, is scored by the probability of its observed counts under dark charge and read noise alone, with a trial factor for its shape:

$$T = -\sum_{i\in g} \ln P(n_i \,|\, \lambda_i, s_i) - \ln N_{\text{shapes}}(m), \qquad P(n \,|\, \lambda, s) = \sum_k \text{Pois}(k;\lambda)\Big[\Phi\big(\tfrac{n+\frac{1}{2}-k}{s}\big) - \Phi\big(\tfrac{n-\frac{1}{2}-k}{s}\big)\Big], \tag{26}$$

with $N_{\text{shapes}}(m)$ the number of 8-connected shapes of m pixels. The group is kept as a "low-E" event when T exceeds the value exceeded by 0.08 groups per Mpix in 10–50 blank images simulated from the image's own background (fixed seed); the analytic bound $\ln(N_{\text{pix}}/F)$ is used only when the blanks have too few groups.

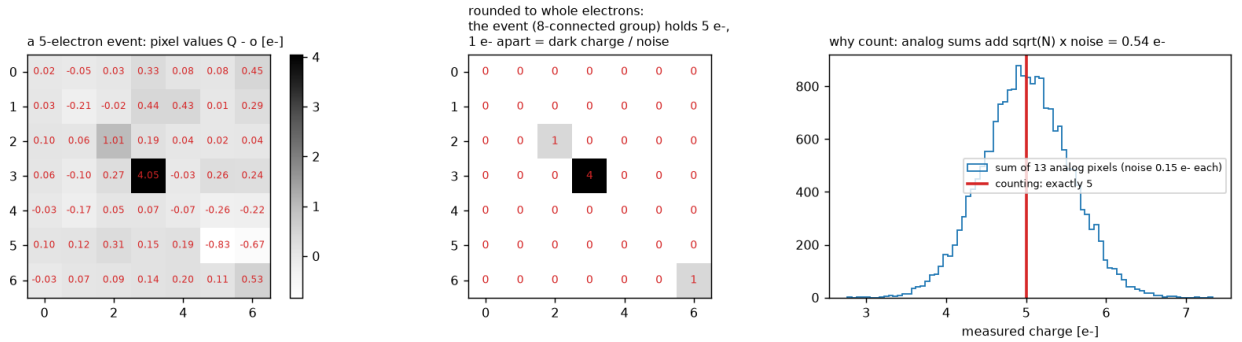


Figure 14: A 5-electron event: analog pixel values, the same rounded to whole electrons (the event's 8-connected group holds $5\,e^-$; the $1\,e^-$ in the corner is a separate dark-charge or noise pixel), and why counting is exact while an analog sum over 13 pixels spreads by $\pm 0.5\,e^-$.

12 Step 8 — classes

Every cluster gets a label from its shape: **low-E** ($E < 0.5$ keV); **x-ray** / **x-ray diffuse** (one round cloud or extent < 6 px; diffuse if $\sigma_{xy} \geq 1$ px); **alpha** (> 1 MeV); **artifact-row / -column** (thin, straight and aligned with a row or column: serial-register streaks, hot columns); **muon** (long, straight skeleton: length ≥ 10 px, scatter of the cloud centres about a line < 1 px and $< 6\,\%$ of the length, *and* at least 2.5 keV per projected pixel — a minimum-ionising particle deposits $\sim$4 keV per 15-µm pixel; a straight track below that is "extended"); **worm** (everything else extended). Labels are only checked on injected objects (alpha 0.94–1.00, x-ray 0.8, muon/worm swap $\sim$30 %).

13 Step 9 — crowded CCDs (crowded mode)

On CCDs covered by tracks (surface runs such as L2 ACM101, the Am-241 test CCD) A_GMX has a *crowded mode* (`clustering_tests/crowded.py`; details, figures and validation in `agmx_crowded.pdf`). It is decided per image from the *core fill*, the fraction of pixels whose 1-px-smoothed charge exceeds $\max(60\,e^-, 5\sigma_\lambda)$: with `crowded='auto'` it runs for core fill ≥ 0.10 (L2 0.27–0.31, Am-241 0.27–0.67, Fe-55 production CCDs 0.015–0.02). Below the threshold A_GMX is unchanged. Crowded mode adds:

- a **background override** for high dark charge (median $\lambda \geq 3\,e^-$/pix, read noise $< 0.45\,e^-$): rows whose whole-electron pedestal is wrong get the good rows' pedestal; per-cell truncated-Poisson dark charge replaces the row model where it differs significantly; sharp-edged hot-column bands are added per column; large faint leftovers are labelled `glow`;
- **serial-register streaks** (one row, no vertical spread, ≥ 8 px) removed before clustering and reported as `artifact-row` clusters with regime `serial`;
- **straight-track hints**: straight tracks are found (Hough on the skeleton of the core, ridge walk through crossings, length, occupancy, width and axis-continuity tests) and only their axis is given to the mixture: after the merge graph and the island arbitration every cloud on a hinted axis goes into one cluster (class `muon`), except elongated clouds crossing it. Width, footprint and charge sharing stay A_GMX's own;
- a **crossing split** of clusters that hold two straight lines crossing as an X;
- **linking** of collinear straight fragments ≥ 12 px;

- by default (`crowded_shapes='on'`, `crowded_shapes.py`) the hinted tracks become **line components** of the mixture (charge per 8-px segment and width linear along the track, fitted jointly with the clouds; every pixel shared by responsibility) and every cluster gets a **topology split** (junctions: the straightest pair stays together; necks between compact deposits; X-ray-like clouds attached through a density dip). Muons carry 8.5 % instead of 28 % foreign charge and their energy loss peaks at the MIP value 0.27 keV/µm.

On L2 injected muons go from 4.1 to 1.9 fragments (median 45 % → 93 % of their charge in one cluster), Fe-55 Kα from 220 to 85 eV width on the worst CCD; on Am-241 the 10–30 keV scale is unchanged (EXT4 $k = 1.0004$ vs 1.0009, EXT3 0.9952 vs 0.9964). Crowded mode is the default (`'auto'`); `crowded='off'` or `AGMX_CROWDED=off` restores the earlier behaviour everywhere.

14 After A_GMX: the few-electron candidate catalogue

A per-event decision at a fixed fake rate cannot do what a rare-event search needs at the lowest dark currents (a 2-e^- single pixel from dark charge is indistinguishable from a real one). For electron-counting images (read noise $< 0.35\,e^-$) `run_AGMX` and `run_WADERs --clusterer agmx` therefore also write `fe_candidates/<image>_ext<n>_fe_candidates.csv` and `.json` (`clustering_tests/fe_candidates.py`), in the style of the low-background experiments (DAMIC-M, SENSEI):

- **candidates**: every group of touching pixels with 2–20 counted electrons (≤ 9 pixels): position, charge, pixels, local dark charge and read noise, the statistic T, n_{exp} = the expected number of dark-charge groups per image at least as unlikely in the clean area (simulated blanks of the image's own occupancy-based dark charge; analytic tail beyond them) and $p_{\mathrm{image}} = 1 - e^{-n_{\mathrm{exp}}}$;
- **mask flags** (nothing is removed): halo within $5\sqrt{E/\mathrm{keV}}$ px (5–60 px) of an A_GMX cluster, charge-transfer trails along its row and column ($E > 1$ keV, 50 px), hot columns, the edge (5 px), overlap with an A_GMX cluster; `clean` = no flag;
- **summary**: the isolated single-electron rate, the clean area, and per class (charge, pixels) the observed clean candidates and the number expected from dark charge alone.

dark charge [e^-/pix]	2 e^- in 1 px	2 e^- in 2 px	3 e^- in 3 px
0.0005	5 / 5.2	58 / 48.2	0 / 0.1
0.001	14 / 12.7	120 / 114.8	0 / 0.7
0.003	89 / 93.4	731 / 734.0	11 / 13.4

Observed / expected clean candidates in 10 simulated blank images (1.7 Mpix each). Injected 3-e^- events are listed in 99 % of cases (median $n_{\mathrm{exp}} = 0.008$ per image). The blanks are generated from the same pixel model the catalogue assumes, so this checks the bookkeeping, not real-detector backgrounds.

15 The worked example

The test scene of figure 2: truth versus both methods (energies in keV).

deposit	truth	A_GMX	WADERs
sharp Fe-55	6.06	6.05 x-ray	6.06
diffuse Fe-55 (σ 1.6 px)	5.81	5.81 x-ray diffuse	5.82
pair, 4 px apart	6.20 + 5.77	**6.19 + 5.78**	11.98 (merged)
diffuse 17.75 keV + sharp Fe-55 3 px away	18.07 + 5.82	**18.13 + 5.78** (sum conserved)	23.97 (merged)
59.5 keV + faint 1.5 keV 6 px away	59.32 + 1.37	59.31 + 1.36	59.35 + 1.36
muon (40 px, minimum ionising)	228.62	228.61 muon	228.79
worm	16.20	16.19 worm	16.24
5-electron event	0.019	not found	0.015

The 5-e^- event (4 e^- in one pixel) sits at $\sim$0.07 e^-/pix dark charge: a group this size is not significant at A_GMX's calibrated fake rate of 0.08 per Mpix there, while WADERs' 30σ seed (3.6 e^-) catches it but has no fake-rate control (section 16.4).

16 Performance

All tests use the same injected objects for every method (same random seeds), plus real lines. The code version of each test is given in its heading.

16.1 The crowded CCD: Am-241 10–30 keV constrained fit (code of 27 October)

29 images, EXT4 (in one corner every pixel holds $> 20\,e^-$), Np L lines with fixed intensity ratios (Lépy et al. 2008), one energy scale k; Z = combined significance of the Np L lines and γ 26.3.

clustering (cut)	clusters	k	dev/ndf	Np Lα	Lβ	Lγ	γ 26.3	Ag K
WADERs ($\sigma_{xy} < 0.7$)	62 665	0.9840	1.74	4.0	15.5	16.0	14.8	12.9
A_GMX ($\sigma_{xy} < 0.7$, blend < 0.02)	73 350	**1.0006**	2.59	5.0	**22.4**	**20.6**	**21.1**	**18.3**
A_GMX (+ ped < 10)	45 464	0.9990	2.12	3.6	17.4	15.9	18.0	15.6

Combined Z: WADERs 27.1, A_GMX 37.4 (29.9 with the pedestal cut). The lines are higher above the continuum and at the right energy. Not understood: the deviance per degree of freedom is higher than WADERs' (2.59 vs 1.74), i.e. the spectrum between the lines is described less well; it rose when the exact pixel model was introduced (1.75 before, on the same images). σ_{xy} is a different width estimate from panaSKImg's, so the 0.7 cut is not the same selection for both.

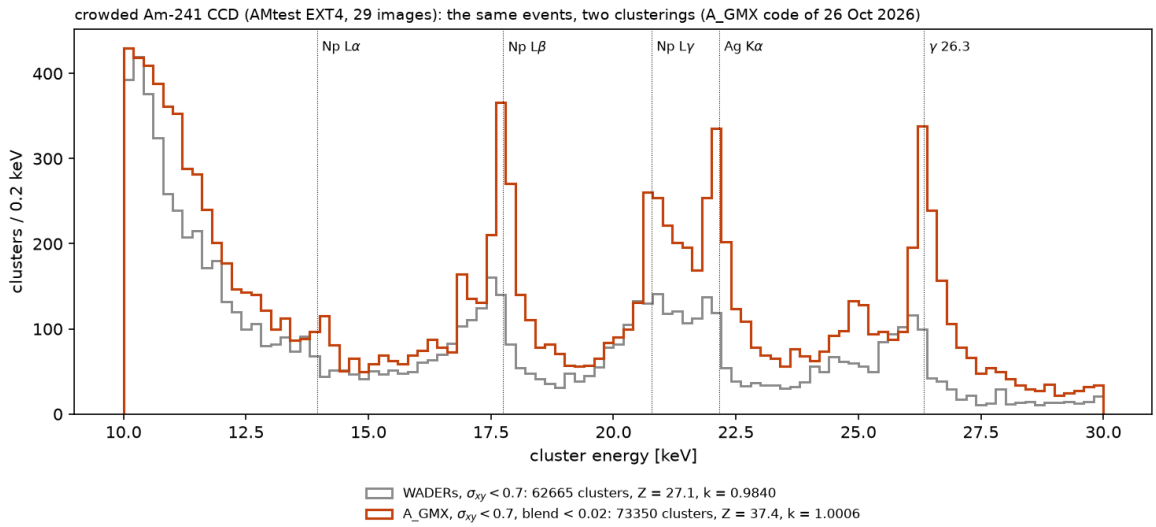


Figure 15: The 10–30-keV spectra of the crowded CCD, WADERs vs A_GMX (blend < 0.02): the same events, the lines about twice as high above the continuum.

16.2 Fe-55 front and back side through the full pipeline (code of 27 October)

DM25 and DM09 (ACM 109), Image_5, 3 images × 4 CCDs each, the full `run_WADERs` chain with the $g(n)$ gain correction, composite-macro definitions (4.5–7 keV; front $\sigma_{xy} < 1$ px, back > 1):

	WADERs nominal	WADERs + $g(n)$	A_GM
DM25 (EXT2–4): Kα back − front	−1.0 %	−2.9 %	**−0.**
DM25: back-side Kα width	112 eV	138 eV	**70**
DM25: back-side rms of the composite profile 3–4 px from the centre	1.10–1.37 px	1.10–1.37 px	1.42–
DM09 (EXT1,3,4): Kα back − front	+1.6 %	+0.1 %	**−0.**
DM09, noisy image (noise 0.30 e^-): back side		−5 to −6.5 %	**−0**

WADERs' back-side loss is charge cut by its thresholds, which scale with the image noise (no loss on DM09's quiet images, 5–6.5 % on its noisy one, ∼3 % on DM25). Nominal WADERs looks better on DM25 only because two errors cancel: panaSKImg's single gain is below the low-charge gain there, so the low-charge pixels of back-side clusters read ∼3 % high. The widths in the A_GMX files include low-charge noise pixels of the footprint; the energies are the cleaner evidence.

16.3 Sharp X-rays next to diffuse ones (simulation, code of 27 October)

Skipper blanks (0.1 e^-/pix, 0.16 e^- noise), 5.9-keV X-rays, 777 scenes of each kind:

scene	cluster	found	$\lvert E/E_{\rm true} - 1\rvert < 5\,\%$
isolated sharp (σ 0.15–0.35)	sharp	1.000	1.000
sharp 3 px from diffuse (σ 1.2–1.6)	sharp / diffuse	0.988 / 0.981	0.967 / 0.967
two sharp 3 px apart	sharp	0.940	0.862
isolated diffuse (σ 1.0–1.6)	diffuse	1.000	1.000

With a Gaussian evaluated at the pixel centre instead of the exact pixel integral, the sharp-on-diffuse case was found in 0.78 and within 5 % in only 0.35 of the scenes.

16.4 Few-electron events at low dark current (code of 26–27 October)

Simulated images (0.15 e^- read noise), 1800 events of 2–5 e^- (Gaussian σ 0.2–0.6 px), 30 blank images per dark charge; full A_GMX:

dark charge [e^-/pix]	3 e^- found	4 e^- found	fakes per 1.7 Mpix
0.0001	0.96	1.00	0.30
0.0005	0.93	0.99	0.37
0.001	0.85	1.00	0.07
0.003	0.40	0.96	0.07
0.01	0.04	0.53	0.23

The remaining fakes at the lowest dark currents are mostly single pixels with 2 e^- (about 0.5 per image at 5×10^{-4} e^-/pix from dark charge and rounding miscounts), indistinguishable from a real 2-e^- event: that is what the candidate catalogue (section 14) is for. WADERs needs $\gtrsim 10\,e^-$ at low dark charge.

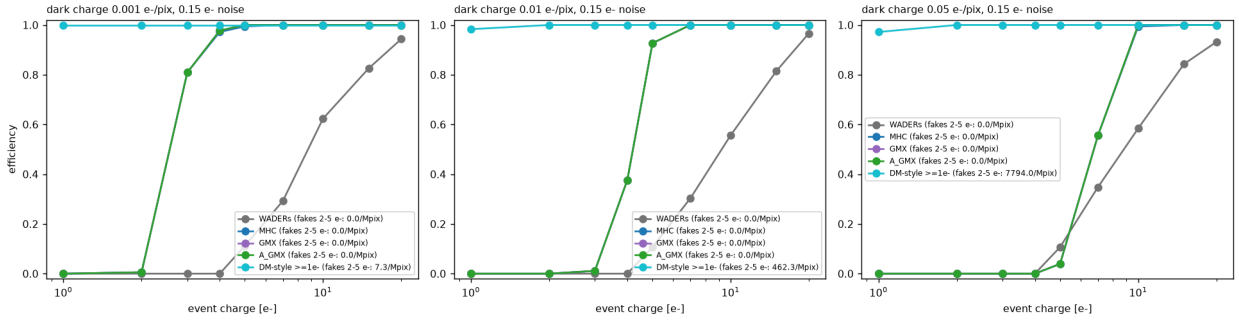


Figure 16: Efficiency against event charge at three dark charges (code of 25 October, before the detection dark charge and the calibrated threshold; the table above is the current code).

16.5 Real data: Fe-55 and Am-241 injections and Kα (code of 27 October)

A subset of the wide test (every 12th Fe-55 image-CCD, every 6th Am-241; 80 image-CCDs), the current pixel model against the Gaussian-at-centre model: Fe-55 injected pairs 4 px apart resolved 0.907 → 0.941, Am-241 0.607 → 0.655; real front-side Kα changed by −0.2 eV (median over image-CCDs); the χ^2 flag of Kα clusters median 6.46 → 1.31.

16.6 The wide test: every image on the machine (code of 25 October)

878 image-CCDs from 16 data sets (Fe-55 production modules DM01–DM28 and PD06–08; Am-241 AMTEst and PD06/DM12; the skip-noise study). Per image-CCD: 12 Fe-55 clusters (σ 0.3–1.6 px), 4 muons, 4 worms, 4 pairs 4 px apart. Success = one cluster with energy within 5 % (pairs: both, within 10 %). "WADERs" here is a hysteresis reimplementation with the WADERs thresholds.

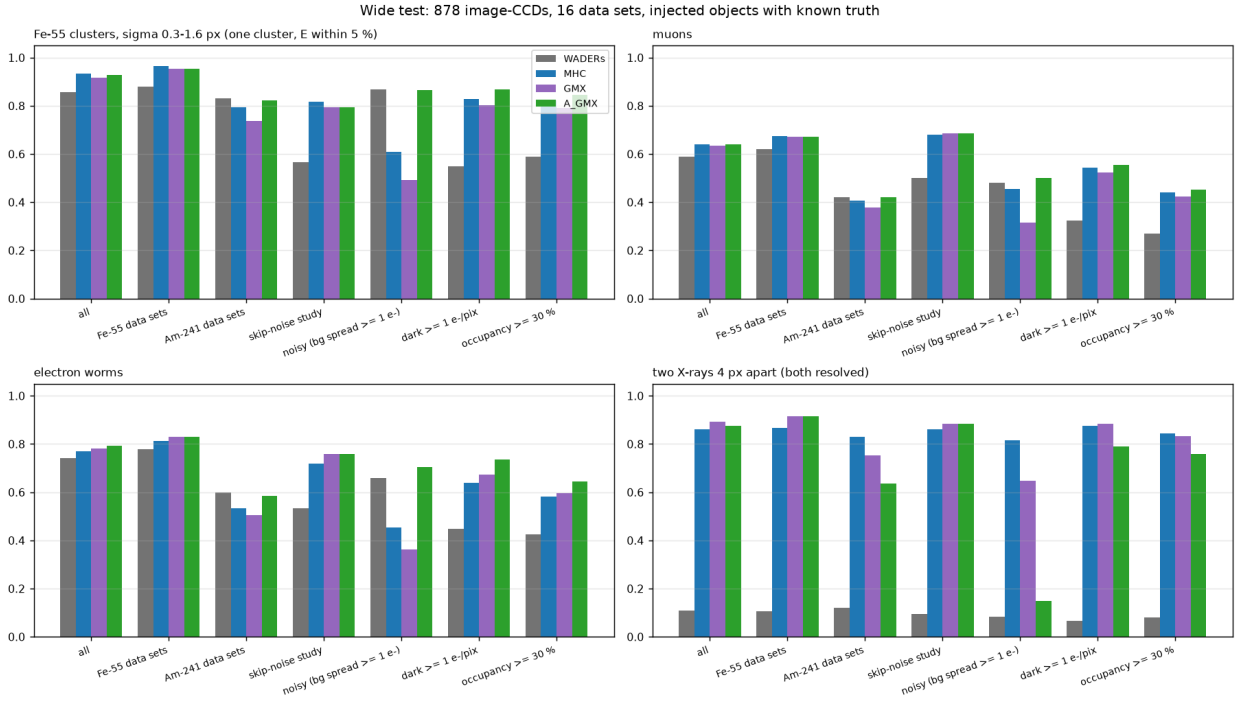


Figure 17: Success fractions by object type and image condition for WADERs, MHC, GMX and A_GMX.

method	Fe-55 ok	Fe-55 E rms	muon	worm	pair 4 px	s / image-CCD
WADERs	0.856	0.52 %	0.59	0.74	0.11	0.01
B auto	0.905	0.59 %	0.62	0.78	0.13	0.01
MFD+G	0.933	0.21 %	0.18	0.37	0.94	0.14
hybrid v1	0.873	0.20 %	0.58	0.66	0.84	0.02
hybrid v2	0.930	0.21 %	0.63	0.68	0.85	0.05
MHC	0.933	0.21 %	0.64	0.77	0.86	1.61
adaptive (MHC engine)	0.931	0.21 %	0.64	0.77	0.83	1.26
GMX	0.916	0.19 %	0.63	0.78	0.89	0.69
A_GMX	**0.927**	**0.19 %**	**0.64**	**0.79**	**0.88**	0.69

Wide test, all 878 image-CCDs.

16.7 Fe-55: energy versus cluster width, pairs, real Kα (code of 25 October)

DM09/DM10 (EXT1, EXT4, images 0–1; 220 injected clusters per width and CCD; pairs of σ 1.4-px clusters).

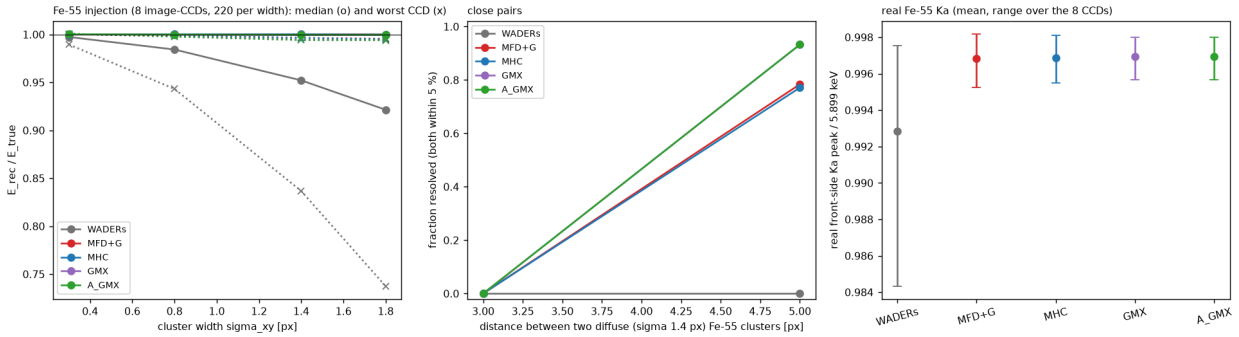


Figure 18: Left: median (circles) and worst-CCD (crosses) energy ratio against cluster width. Middle: diffuse pairs resolved. Right: real front-side Kα over the 8 CCDs.

method	σ 0.3	0.8	1.4	1.8 px	pair 5 px	real Kα/5.899	spread
WADERs	0.9974 (0.990)	0.9843 (0.943)	0.9521 (0.837)	0.9214 (0.737)	0.00	0.9928	1.32 %
MFD+G	1.0001 (1.000)	1.0002 (0.999)	1.0001 (0.996)	0.9999 (0.995)	0.78	0.9968	0.29 %
MHC	1.0001 (1.000)	1.0002 (0.999)	1.0001 (0.996)	0.9998 (0.995)	0.77	0.9969	0.26 %
A_GMX	1.0001 (1.001)	0.9999 (0.998)	0.9996 (0.994)	0.9997 (0.994)	**0.93**	0.9969	**0.23 %**

Median $E_{\mathrm{rec}}/E_{\mathrm{true}}$ (worst CCD in brackets) per cluster width; pairs resolved; real Kα.

16.8 Injected tracks, alphas and artifacts in real Am-241 images (code of 25 October)

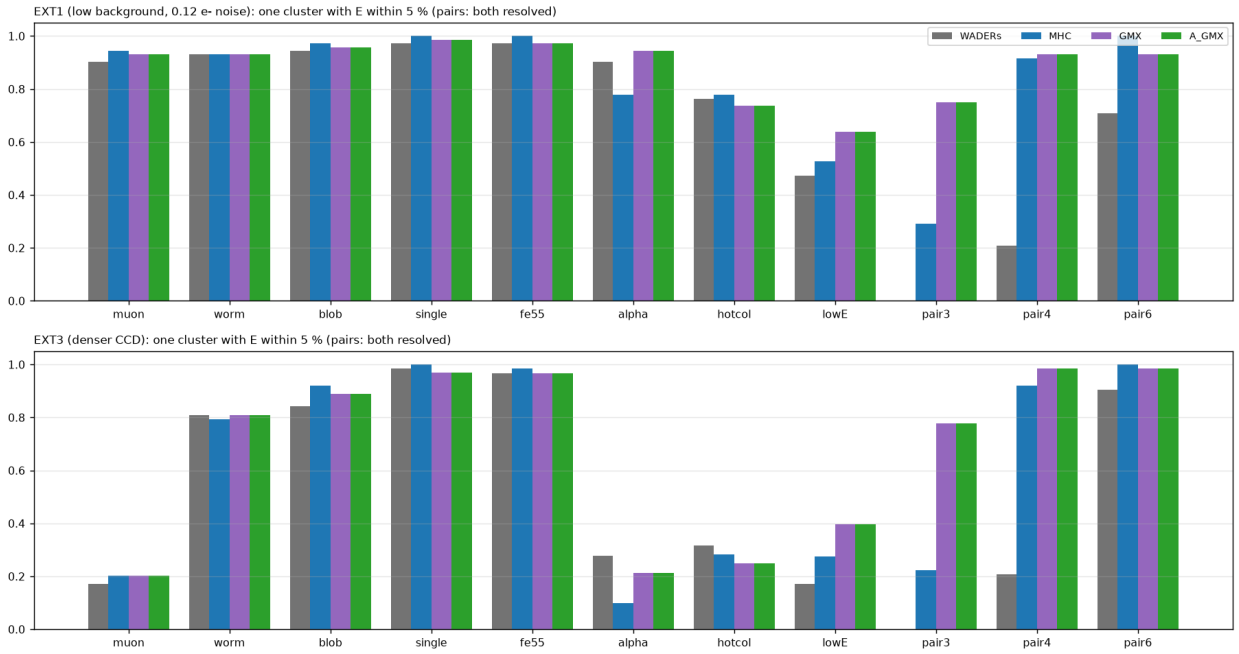


Figure 19: EXT1 (low background) and EXT3 (denser CCD): success per object type.

method	muon	worm	blob	single	fe55	alpha	hotcol	lowE	pair3	pair4	pair6
WADERs	0.90	0.93	0.94	0.97	0.97	0.90	0.76	0.47	0.00	0.21	0.71
MFD+G	0.19	0.28	0.97	1.00	1.00	0.78	0.00	0.47	0.29	0.99	1.00
MHC	0.94	0.93	0.97	1.00	1.00	0.78	0.78	0.53	0.29	0.92	1.00
A_GMX	0.93	0.93	0.96	0.99	0.97	**0.94**	0.74	**0.64**	**0.75**	0.93	0.93

EXT1, 3 images × 3 seeds: success fraction per object type. The injected muons were fainter than minimum ionising (0.94 keV per pixel); the harness now has physical muons (`MUON_PHYS=1`) for the next run.

16.9 Run time

Median per image-CCD: Fe-55 0.9 s, Am-241 62 s (code of 27 October; WADERs 0.01 s). Crowded and high-dark-charge CCDs take minutes: many seeds and clouds, and the few-electron tests have far more candidate groups.

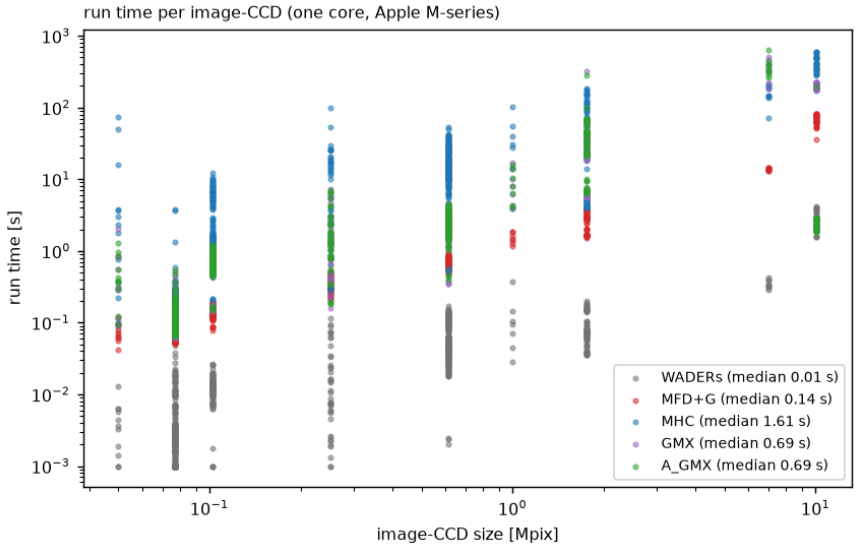


Figure 20: Run time per image-CCD against its size (code of 25 October; one core, measured with 13 runs in parallel, so times are indicative).

17 Quality flags and the selection for X-ray lines

On the crowded Am-241 CCD the 59.5 keV line has a high-side tail. Injected X-rays (5.9–59.5 keV) at random positions are unbiased away from real events (median $E_{\mathrm{rec}}/E_{\mathrm{true}} = 0.9998$, spread 0.03 % at 59.5 keV); next to real events every method, WADERs included, adds charge of the neighbour. Two mechanisms, both real charge of other deposits:

1. **a charge sheet under the cluster** (dominant on the crowded CCD): many small overlapping deposits make a diffuse sheet of 20–200 e^-/pix; sharing gives the part under a cluster's footprint to the cluster, which still looks clean. The median charge in a ring 5–8 px around 59.5-keV-like clusters is 24 e^-/pix at the peak but 116–218 e^-/pix at 60.5–66 keV;
2. **a neighbouring deposit merged into the cluster**: such clusters have several clouds and a bad fit of their own model.

Re-measuring the energy (clipping, local backgrounds, core apertures, a local pedestal subtracted over the footprint) either changes nothing or adds so much noise that the lines broaden. Energies are therefore left unchanged and the flags of step 6 select clean clusters.

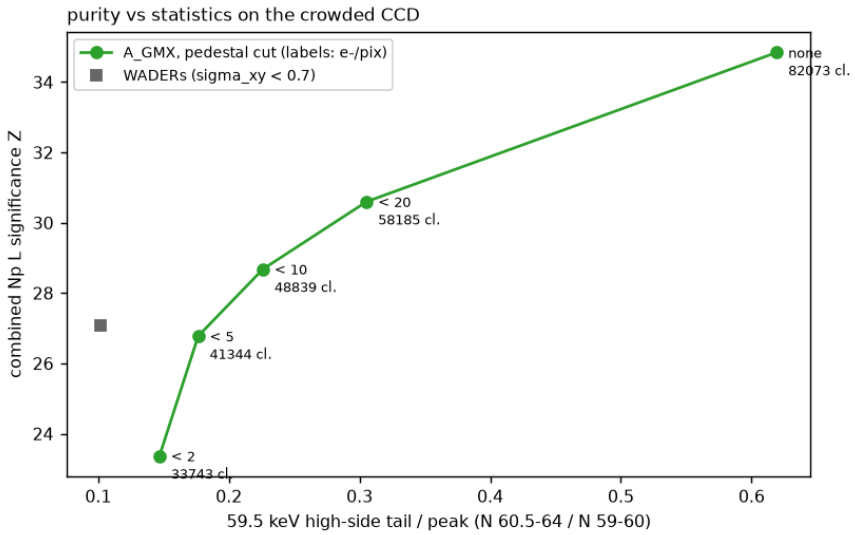


Figure 21: Purity against statistics: the 59.5 keV tail and the Np L significance for pedestal cuts of 20, 10, 5, 2 e^-/pix (code of 25 October).

Recommended selection for spectroscopy. $\sigma_{xy} < 0.7$ (Am-241) or < 1 (front-side Fe-55), blend < 0.02, ped $< 10\,e^-$/pix, not on the edge; add peakfrac > 0.95 where merged neighbours matter (crowded fields; not for Fe-55, where it removes 28–41 % of the front-side Kα clusters). On Fe-55 images, which have no charge sheet, the ped cut removes at most 0.1 % of the front-side Kα clusters. The tables below (code of 25 October) give every Am-241 line fitted alone and the front-side Fe-55 Kα of every data set, with and without the flags.

selection (crowded CCD, 29 images)	clusters	59.5 tail/peak	k	dev/ndf	Z	Lβ / Lγ / γ26.3 / Ag K
WADERs ($\sigma_{xy} < 0.7$)	62 665	0.10	0.9840	1.74	27.1	15.5 / 16.0 / 14.8 / 12.9
A_GMX ($\sigma_{xy} < 0.7$, blend < 0.02)	82 073	0.62	1.0016	1.93	34.8	20.8 / 19.2 / 19.9 / 18.1
+ ped $< 10\,e^-$/pix	48 839	0.23	0.9994	1.55	28.7	16.2 / 16.0 / 17.3 / 15.3
+ ped $< 5\,e^-$/pix	41 344	0.18	0.9993	1.49	26.8	14.6 / 15.3 / 16.3 / 14.3

CCD	selection	clusters	Np La	Np Lb	Np Lg	γ 26.3	γ 59.5	59.5 tail	k (EXT4)
EXT1	old A-GMX (sxy<0.7, blend<0.02)	13632	–	–	–	–	0.9867 (206)	0.23	
EXT1	new A-GMX (sxy<0.7, blend<0.02)	13715	–	–	–	0.9716 (251)	0.9867 (206)	0.23	
EXT1	new + ped<10	13715	–	–	–	0.9716 (251)	0.9867 (206)	0.23	
EXT1	new + peakfrac>0.95	13600	–	–	–	0.9723 (247)	0.9867 (208)	0.23	
EXT1	new + ped<10 + peakfrac>0.95	13600	–	–	–	0.9723 (247)	0.9867 (208)	0.23	
EXT3	old A-GMX (sxy<0.7, blend<0.02)	75580	–	0.9943 (200)	–	–	0.9989 (246)	0.11	
EXT3	new A-GMX (sxy<0.7, blend<0.02)	83037	–	–	–	–	0.9989 (246)	0.11	
EXT3	new + ped<10	81364	–	0.9953 (200)	–	–	0.9988 (242)	0.10	
EXT3	new + peakfrac>0.95	65083	–	0.9962 (200)	–	–	0.9994 (244)	0.14	
EXT3	new + ped<10 + peakfrac>0.95	63641	–	0.9962 (200)	–	–	0.9994 (239)	0.13	
EXT4	old A-GMX (sxy<0.7, blend<0.02)	65876	–	–	–	1.0038 (273)	1.0017 (353)	0.60	1.0014
EXT4	new A-GMX (sxy<0.7, blend<0.02)	82073	–	–	–	1.0041 (288)	1.0017 (352)	0.62	1.0016
EXT4	new + ped<10	48839	–	0.9977 (200)	–	1.0007 (245)	1.0006 (314)	0.23	0.9994
EXT4	new + peakfrac>0.95	70840	–	–	1.0053 (263)	1.0046 (296)	1.0025 (349)	0.69	1.0018
EXT4	new + ped<10 + peakfrac>0.95	41474	–	0.9983 (200)	–	1.0007 (250)	1.0012 (306)	0.24	0.9993

Every Am-241 line fitted alone: peak / true energy (width in eV); Np Lβ and Lγ are groups of lines, so their centres are not expected at exactly 17.75 / 20.78 keV (the constrained fit's k is the check there).

data set	n	Ka old (all)	Ka new (all)	front	front+ped	front+pf	front+both	front lost to ped	front lost to both
2025-08-27-DM07 DM09 DM10	102	5.8004	5.8000	5.7705 (63)	5.7705 (63)	5.7772 (62)	5.7772 (62)	0.0 %	28.7 %
2025-08-30-DM11 DM13 DM14	102	5.7960	5.7955	5.7732 (68)	5.7732 (68)	5.7915 (65)	5.7915 (65)	0.0 %	40.5 %
2025-09-03-DM15 DM16 DM17	102	5.7794	5.7789	5.7525 (67)	5.7518 (67)	5.7721 (65)	5.7721 (65)	0.1 %	37.4 %
2025-09-06-DM18 DM19 DM20	102	5.7719	5.7715	5.7387 (70)	5.7387 (70)	5.7524 (69)	5.7524 (69)	0.0 %	33.6 %
2025-09-13-DM24 DM25 DM26	90	5.9069	5.9070	5.8875 (65)	5.8875 (65)	5.8895 (66)	5.8895 (66)	0.0 %	38.0 %

Fe-55 K-alpha per data set: median fitted peak over image-CCDs [keV] (median width eV); old vs new A_GMX without cuts, new front side with the flag cuts. No K-alpha fit in any image-CCD of: 2025-09-07-DM18 DM19 DM20, 2025-09-09-DM21 DM22 DM23, 2025-09-16-DM08 DM11 DM12, 2025-09-19-DM07 DM27 DM28, 2025-10-31-DM04 DM05 DM06, 2025-11-04-DM10 DM24 DM25, 2025-11-08-PD06 PD07 PD08, 2025-11-13-DM01 DM02 DM03.

selection	image-CCDs	median K-alpha [keV]	spread [eV]	width [eV]
old A_GMX, all clusters	401	5.7934	66	79
new A_GMX, all clusters	401	5.7947	67	80
new, front	381	5.7709	69	67
new, front, ped<10	380	5.7704	69	67
new, front, peakfrac>0.95	343	5.7851	73	66
new, front, ped<10, peakfrac>0.95	343	5.7851	73	66

Fe-55 K-alpha, all Fe-55 image-CCDs: number fitted, median peak, robust spread over image-CCDs, median width. old = A_GMX before the 25 October changes, new = with them (satellite limit, flags, audit fixes).

data sets	image-CCDs	Fe-55	muon	worm	pair 4 px
Fe-55	705	0.951 / 0.954	0.672 / 0.672	0.829 / 0.829	0.912 / 0.915
Am-241	115	0.819 / 0.824	0.424 / 0.422	0.585 / 0.585	0.622 / 0.635
skip-noise	58	0.799 / 0.792	0.675 / 0.680	0.780 / 0.775	0.899 / 0.888

Injection test (same objects): success old / new A_GMX.

18 Does A_GMX supersede the other methods?

Yes, as the general clusterer. Against each alternative:

method	where it is still better than A_GMX	verdict
MHC	Fe-55 data sets: Fe-55 success 0.964 vs 0.951 (energies equal); hot columns 0.78 vs 0.74	**superseded.** A_GMX is better on crowded images, noisy images, high dark charge, worms, alphas, 3-px pairs (0.75 vs 0.29), diffuse 5-px pairs (0.93 vs 0.77) and energy resolution. Keep the code as a cross-check.
hybrid v1 / v2, adaptive (MHC engine), GMX alone	nothing significant	**superseded**.
MFD+G	isolated X-rays only: Fe-55 success 0.972 vs 0.951, faster	**keep for fast Fe-55 gain calibration**. Not for tracks (muon 0.18).
WADERs	noisy "unmodelled" images (where A_GMX *is* WADERs-like); speed ($\sim$100$\times$); the established pipeline; the smoothness of the crowded-CCD spectrum between the lines	**keep as the reference**; it is what A_GMX falls back to.

19 Limits and open items

- **Crowded-CCD fit quality:** deviance/ndf 2.59 vs 1.74 (WADERs); not understood.
- **Background on crowded CCDs:** the fitted dark charge on the crowded CCD is $0.157\,e^-$/pix (0.093 with the background model of 24 October), which moves the 59.5-keV peak by about +0.1 keV; which value is right is being tested.
- **Unmodelled images** fall back to hysteresis: close pairs are then not resolved; the pixel threshold of $4.5\,e^-$ is a stop-gap.
- **Very sharp clusters:** widths below ≈ 0.25 px are hard to measure.
- **EM depends on the seeds**; no global optimum is guaranteed. The island test tolerates 5 % non-Gaussianity and has only been checked on injected objects and Fe-55; real Am-241 tracks have not been inspected by eye.
- **Labels:** muon/worm swap $\sim$30 %; classes only validated on injections.
- **Speed:** minutes on crowded and high-dark-charge CCDs.
- **Crowded mode** (section 13): straight stretches of electron tracks are taken as straight tracks; 59.5 keV loses 2.8 % of its events on Am-241 EXT3.
- **Validation:** the wide, Fe-55-width and track tests are from the code of 25 October; injections use Gaussian clouds, which is A_GMX's own model; the few-electron blanks are generated from the pixel model the finder assumes.

20 Parameters, code, how to run

parameter	default	where	meaning
`excess_ratio`, `noise_max`	3, $1\,e^-$	adaptive	unmodelled if $s \geq 3\sigma_{\rm read}$ and $\geq 0.5\,e^-$, or $\sigma_{\rm read} \geq 1\,e^-$
`occ_thr`, `box`	0.3, 31 px	adaptive	"crowded" label (information)
`scales`	0.5, 1, 1.5, 2.5 px	GMX	matched-filter widths
`nsig`	5	GMX	seed significance (+ Poisson check at the same tail)
`n_iter`	30 (+15 residual, +5 pruning)	GMX	EM iterations
`nsig_res`	5	GMX	residual-pass seeds
`sig_min`	4	GMX	prune clouds with S/N below
`r_merge`, `w_ratio`	0.5, 1.6	GMX	ridge ratio, width ratio for merging
`f_bridge`, `a_faint`	0.1, $50\,e^-$	GMX	fragments link only to their brightest partner
`link_alpha`, `island_max`, `bic_margin`	0.01, 4000 px, 100	GMX	island test
`f_sat`, `sat_e_max`	0.01, 1 keV	GMX	satellite absorption (only fragments < 1 keV)
`ped_ring`, `ped_q`	(5, 8) px, 25	GMX	local pedestal flag (ring, percentile)
`grow`	3 px	GMX	CTI-trail growth
`count_fakes`, `count_E_max`	0.08 per Mpix, $100\,e^-$	GMX	few-electron finder, counting energies

Switches (environment): `AGMX_PIX_EXACT=0` (Gaussian-at-centre model, island margin 30, no fragment rule), `AGMX_PHYS_FIXES=0` (the background, hot-column, counting, width and class rules of before 25 October), `AGMX_NUMBA=0` (numpy EM), `AGMX_FE_CANDIDATES=0` (no candidate catalogue), `AGMX_CROWDED_SHAPES=on|off` (muon lines + topology in crowded mode; default `on`), `AGMX_CROWDED=auto|on|off` (crowded mode, section 13; default `auto`; also `run_WADERs/run_AGMX --agmx-crowded`).

```
import sys; sys.path.insert(0, "comm_scripts/clustering_tests")
import adaptive as AD
clusters = AD.adaptive_cluster(Q_electrons, valid_mask, engine="gmx")
# each cluster: E [keV], y, x, sxy [px], cls, blend, sig, n_comp, edge, ped [e-/pix],
    peakfrac, chi2, saturated,
# regime ('sparse' / 'crowded' / 'unmodelled'), occ
good = [c for c in clusters if c.get("blend", 0) < 0.02 and not c.get("edge")
       and not c.get("ped", 0) >= 10 # not sitting on a charge sheet (crowded fields)
       and c.get("peakfrac", 1) > 0.95] # not merged with a neighbouring deposit (not for
            Fe-55)
```

Test scripts (all in `clustering_tests/`, README there): `wide_test.py` + `wide_run.py adaptive+GMX` (all images), `adaptive_validate.py` (tracks, few-electron, Fe-55, crowded), `make_agmx_figures.py`, `make_agmx_perf.py`. Unit tests: `python -m unittest discover -s tests`.

20.1 Running A_GMX on WADERs data: cluster files in the WADERs format

```
run_WADERs images.txt --gain-correction --clusterer agmx --composite # A_GMX files,
    composite on them
run_WADERs images.txt --gain-correction --clusterer both --composite # also
    composite_agmx/ next to composite/
run_AGMX WADERs_analysis_<run> [-j 4] [--per-image] [--no-gaincorr] [--no-fe-candidates]
     # a finished run
```

It runs on the calibrated image panaSKImg saves for every CCD (`<image>_EXT<n>_waders.fits.bz2`). With the gain correction, the image is first converted ADU $\rightarrow e^-$ with the same $g(n)$ curves and anchor as `root_gaincorr/`; a CCD without a reliable curve keeps its fitted gain, as there. The output is `root_agmx/` and `root_agmx_gaincorr/`: one `panaSKImg_clustersRec_<image>_ext<n>.root` per CCD, written by `src/agmx_root.py`. Every A_GMX cluster (its pixels and the charge it takes from each, which sums to E) becomes

a pysimdamicm `Cluster` object, the ClusterFinder's own class, so all branches (positions, `wSTD`, `fSTD`, PCA, track end points, seed, `closest_cls_*`, `Energy_PS/AVG`, …) are computed by panaSKImg's code. `info` and `process_config` are copied from the CCD's panaSKImg file, and the extra branches `agmx_cls`, `agmx_blend`, `agmx_ped`, `agmx_peakfrac`, `agmx_chi2`, `agmx_sig`, `agmx_regime` and a `TNamed agmx_config` are added. MakeComposite and every other reader of WADERs files read them unchanged. For electron-counting CCDs the candidate catalogue goes to `fe_candidates/`.

Check. `clustering_tests/agmx_root_check.py` writes the clusters of real panaSKImg files (EXT1, EXT3 and EXT4 of an Am-241 image: 1652, 7686 and 5452 clusters) through the writer and compares every branch. They are identical except in numerically degenerate cases, where panaSKImg's own result depends on the last bits of its inputs (Gaussian fits of profiles 1–3 bins wide, the sign of a PCA eigenvector, PCA of 1–2 pixel clusters, `closest_cls_*` when the nearest cluster lies in the same row or column). With the default `--clusterer waders` the pipeline output is identical to before. A flag value of −999 in the files means "not computed" (e.g. `agmx_chi2` of a cluster the mixture did not fit); clusters from the few-electron finder and the hysteresis branch get `edge` and `ped` like the others, `blend` = 0 and `peakfrac` = 1.

A All performance tables

Generated by `clustering_tests/make_agmx_perf.py` from the saved results (code of 25 October; same tests as section 16; all methods; "ok" = one cluster with energy within 5 %, pairs: both resolved).

method	Fe-55 ok	Fe-55 E rms	muon	worm	pair 4 px	s / image-CCD (median)
WADERs	0.856	0.52 %	0.59	0.74	0.11	0.01
B auto	0.905	0.59 %	0.62	0.78	0.13	0.01
MFD+G	0.933	0.21 %	0.18	0.37	0.94	0.14
hybrid v1	0.873	0.20 %	0.58	0.66	0.84	0.02
hybrid v2	0.930	0.21 %	0.63	0.68	0.85	0.05
MHC	0.933	0.21 %	0.64	0.77	0.86	1.61
adaptive (MHC engine)	0.931	0.21 %	0.64	0.77	0.83	1.26
GMX	0.916	0.19 %	0.63	0.78	0.89	0.69
A_GMX	0.927	0.19 %	0.64	0.79	0.88	0.69

Wide test, all (878 image-CCDs)

method	Fe-55 ok	Fe-55 E rms	muon	worm	pair 4 px	s / image-CCD (median)
WADERs	0.880	0.47 %	0.62	0.78	0.11	0.01
B auto	0.927	0.53 %	0.65	0.81	0.13	0.01
MFD+G	0.972	0.17 %	0.20	0.39	0.95	0.13
hybrid v1	0.926	0.16 %	0.63	0.71	0.87	0.02
hybrid v2	0.972	0.17 %	0.68	0.73	0.88	0.04
MHC	0.964	0.17 %	0.68	0.81	0.87	0.94
adaptive (MHC engine)	0.961	0.17 %	0.67	0.81	0.87	0.83
GMX	0.954	0.16 %	0.67	0.83	0.91	0.59
A_GMX	0.954	0.16 %	0.67	0.83	0.91	0.60

Wide test, Fe-55 data sets (705 image-CCDs)

method	Fe-55 ok	Fe-55 E rms	muon	worm	pair 4 px	s / image-CCD (median)
WADERs	0.830	0.77 %	0.42	0.60	0.12	0.11
B auto	0.822	0.89 %	0.42	0.60	0.14	0.24
MFD+G	0.738	1.21 %	0.11	0.27	0.88	3.79
hybrid v1	0.605	0.78 %	0.29	0.37	0.69	8.10
hybrid v2	0.728	1.18 %	0.37	0.46	0.72	16.89
MHC	0.793	1.11 %	0.41	0.53	0.83	76.69
adaptive (MHC engine)	0.794	0.95 %	0.41	0.56	0.61	38.31
GMX	0.736	0.75 %	0.38	0.50	0.75	40.84
A_GMX	0.824	0.72 %	0.42	0.58	0.63	23.31

Wide test, Am-241 data sets (115 image-CCDs)

method	Fe-55 ok	Fe-55 E rms	muon	worm	pair 4 px	s / image-CCD (median)
WADERs	0.567	1.59 %	0.50	0.53	0.10	0.01
B auto	0.784	1.58 %	0.64	0.76	0.18	0.03
MFD+G	0.824	1.18 %	0.14	0.29	0.97	0.32
hybrid v1	0.738	1.05 %	0.52	0.60	0.79	0.07
hybrid v2	0.804	1.17 %	0.55	0.58	0.81	0.21
MHC	0.817	1.17 %	0.68	0.72	0.86	10.70
adaptive (MHC engine)	0.812	1.17 %	0.68	0.71	0.85	7.77
GMX	0.794	1.17 %	0.69	0.76	0.88	1.30
A_GMX	0.794	1.17 %	0.69	0.76	0.88	1.30

Wide test, skip-noise study (58 image-CCDs)

method	Fe-55 ok	Fe-55 E rms	muon	worm	pair 4 px	s / image-CCD (median)
WADERs	0.867	1.66 %	0.48	0.66	0.08	2.40
B auto	0.802	2.08 %	0.39	0.65	0.15	2.46
MFD+G	0.281	9.15 %	0.08	0.09	0.73	69.85
hybrid v1	0.275	8.24 %	0.13	0.10	0.67	98.38
hybrid v2	0.278	8.99 %	0.15	0.10	0.68	156.01
MHC	0.608	5.62 %	0.45	0.45	0.81	396.59
adaptive (MHC engine)	0.864	2.02 %	0.50	0.70	0.15	1.29
GMX	0.491	4.42 %	0.31	0.36	0.65	193.75
A_GMX	0.864	2.02 %	0.50	0.70	0.15	2.38

Wide test, noisy (bg spread $\geq$ 1 e-) (27 image-CCDs)

method	Fe-55 ok	Fe-55 E rms	muon	worm	pair 4 px	s / image-CCD (median)
WADERs	0.550	2.40 %	0.33	0.45	0.07	0.08
B auto	0.806	2.10 %	0.48	0.66	0.17	0.08
MFD+G	0.772	1.07 %	0.15	0.29	0.91	0.70
hybrid v1	0.724	1.02 %	0.39	0.51	0.82	0.11
hybrid v2	0.765	1.06 %	0.44	0.52	0.83	0.32
MHC	0.829	1.00 %	0.54	0.64	0.88	16.18
adaptive (MHC engine)	0.868	0.94 %	0.55	0.68	0.75	7.31
GMX	0.803	0.88 %	0.52	0.67	0.88	2.05
A_GMX	0.869	0.90 %	0.56	0.74	0.79	1.97

Wide test, dark ≥ 1 e-/pix (152 image-CCDs)

method	Fe-55 ok	Fe-55 E rms	muon	worm	pair 4 px	s / image-CCD (median)
WADERs	0.590	1.64 %	0.27	0.43	0.08	0.08
B auto	0.781	1.60 %	0.39	0.58	0.15	0.18
MFD+G	0.789	1.05 %	0.12	0.29	0.90	1.55
hybrid v1	0.684	0.84 %	0.32	0.44	0.74	0.36
hybrid v2	0.781	1.02 %	0.39	0.50	0.77	2.43
MHC	0.822	0.98 %	0.44	0.58	0.85	25.13
adaptive (MHC engine)	0.819	0.92 %	0.44	0.59	0.70	11.35
GMX	0.791	0.79 %	0.42	0.60	0.83	9.08
A_GMX	0.845	0.77 %	0.45	0.65	0.76	2.50

Wide test, occupancy ≥ 30 % (194 image-CCDs)

method	sigma 0.3	0.8	1.4	1.8	pair 5 px resolved	real Ka / 5.899	Ka spread
WADERs	0.9974 (0.990)	0.9843 (0.943)	0.9521 (0.837)	0.9214 (0.737)	0.00	0.9928	1.32 %
MFD+G	1.0001 (1.000)	1.0002 (0.999)	1.0001 (0.996)	0.9999 (0.995)	0.78	0.9968	0.29 %
MHC	1.0001 (1.000)	1.0002 (0.999)	1.0001 (0.996)	0.9998 (0.995)	0.77	0.9969	0.26 %
GMX	1.0001 (1.001)	0.9999 (0.998)	0.9996 (0.994)	0.9997 (0.994)	0.93	0.9969	0.23 %
A_GMX	1.0001 (1.001)	0.9999 (0.998)	0.9996 (0.994)	0.9997 (0.994)	0.93	0.9969	0.23 %

Fe-55 injection (DM09 / DM10, EXT1 / EXT4, images 0-1): median E_rec/E_true (worst CCD) per width, pairs, real Ka

method	muon	worm	blob	single	fe55	alpha	hotcol	lowE	pair3	pair4	pair6
WADERs	0.90	0.93	0.94	0.97	0.97	0.90	0.76	0.47	0.00	0.21	0.71
MFD+G	0.19	0.28	0.97	1.00	1.00	0.78	0.00	0.47	0.29	0.99	1.00
MHC	0.94	0.93	0.97	1.00	1.00	0.78	0.78	0.53	0.29	0.92	1.00
GMX	0.93	0.93	0.96	0.99	0.97	0.94	0.74	0.64	0.75	0.93	0.93
A_GMX	0.93	0.93	0.96	0.99	0.97	0.94	0.74	0.64	0.75	0.93	0.93

Injected objects in real Am-241 images, EXT1 (3 images x 3 seeds, success fraction)

method	muon	worm	blob	single	fe55	alpha	hotcol	lowE	pair3	pair4	pair6
WADERs	0.17	0.81	0.84	0.98	0.97	0.28	0.32	0.17	0.00	0.21	0.90
MFD+G	0.02	0.24	0.92	1.00	0.98	0.10	0.02	0.34	0.22	0.98	1.00
MHC	0.20	0.79	0.92	1.00	0.98	0.10	0.28	0.28	0.22	0.92	1.00
GMX	0.20	0.81	0.89	0.97	0.97	0.21	0.25	0.40	0.78	0.98	0.98
A_GMX	0.20	0.81	0.89	0.97	0.97	0.21	0.25	0.40	0.78	0.98	0.98

Injected objects in real Am-241 images, EXT3 (3 images x 3 seeds, success fraction)

dark e-/pix	method	2 e-	3 e-	4 e-	5 e-	7 e-	fakes/Mpix
0.001	WADERs	0.00	0.00	0.00	0.11	0.29	0.0
0.001	MHC	0.01	0.81	0.97	0.99	1.00	0.0
0.001	GMX	0.01	0.81	0.98	1.00	1.00	0.0
0.001	A_GMX	0.01	0.81	0.98	1.00	1.00	0.0
0.001	DM-style ≥1e-	1.00	1.00	1.00	1.00	1.00	7.3
0.01	WADERs	0.00	0.00	0.00	0.11	0.30	0.0
0.01	MHC	0.00	0.01	0.38	0.93	1.00	0.0
0.01	GMX	0.00	0.01	0.38	0.93	1.00	0.0
0.01	A_GMX	0.00	0.01	0.38	0.93	1.00	0.0
0.01	DM-style ≥1e-	1.00	1.00	1.00	1.00	1.00	462.3
0.05	WADERs	0.00	0.00	0.00	0.11	0.35	0.0
0.05	MHC	0.00	0.00	0.00	0.04	0.56	0.0
0.05	GMX	0.00	0.00	0.00	0.04	0.56	0.0
0.05	A_GMX	0.00	0.00	0.00	0.04	0.56	0.0
0.05	DM-style ≥1e-	1.00	1.00	1.00	1.00	1.00	7794.0

Few-electron events on simulated blanks (efficiency by event charge; fakes = clusters of 2-5 e- per Mpix with nothing injected)

clusters	N	k	dev/ndf	Np La	Np Lb	Np Lg	gamma 26.3	Ag K	Z (Np L + 26.3)
WADERs (sigma_xy < 0.7)	62665	0.9840 ± 0.0008	1.74	4.0	15.5	16.0	14.8	12.9	27.1
A_GMX (sigma_xy < 0.7, blend < 0.02)	82073	1.0016 ± 0.0004	1.93	3.7	20.8	19.2	19.9	18.1	34.8

Crowded Am-241 CCD (runs_29img EXT4, 29 images): constrained 10-30 keV fit, line significances [sigma]

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